Univariate Data Analysis

Gongqiu Zhang

SSE, CUHK-Shenzhen

Outline

1 Introduction to Univariate Data Analysis

2 Random Variables and Distributions

3 Estimation of Risk Measures

4 Extraction of Risk-Neutral Distributions from Option Data

Univariate Data Analysis

Sometimes, we are not interested in the relationship between variables, but only in the distribution of a single variable or even some of its distributional characteristics.

Examples:

- The distribution of the daily returns of a stock/index.
- The distribution of the daily returns of a trading strategy.

In such cases, we often regard the daily returns as independent and identically distributed samples from the same distribution, or say the daily returns are realizations of a random variable.

The goal of univariate data analysis is to understand the distribution of the random variable.

Simple and Log Returns

Log return/continuously compounded return:

$$r_t = \log\left(S_t/S_{t-1}\right),$$

where S_t is the price (adjusted for dividends) at day t.

Simple return:

$$R_t = S_t/S_{t-1} - 1 \approx (\geq) r_t, \quad \text{for small } r_t.$$

We use log return more often because:

- Log return is additive in the sense that the log return over a period is the sum of the log returns over the subperiods.
- Simple returns are distributed in $[-1, \infty)$, while log returns are distributed in $(-\infty, \infty)$.

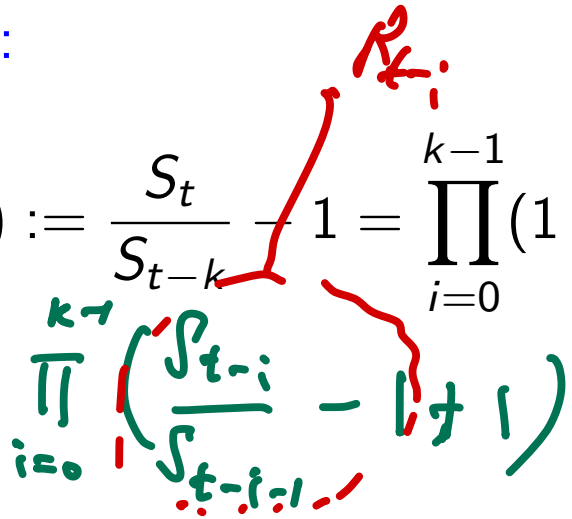
k-Period Returns

One-period gross return:

$$G_t := \frac{S_t}{S_{t-1}}.$$

k-period simple return:

$$R_t(k) := \frac{S_t}{S_{t-k}} - 1 = \prod_{i=0}^{k-1}(1 + R_{t-i}) - 1.$$



k-period log return:

$$r_t(k) := \ln\left(\frac{S_t}{S_{t-k}}\right) = \sum_{i=0}^{k-1} r_{t-i}.$$

The Daily Returns of SPX

The daily returns of S&P 500 index from 2000-12-18 to 2024-12-05. Data source: Yahoo Finance.

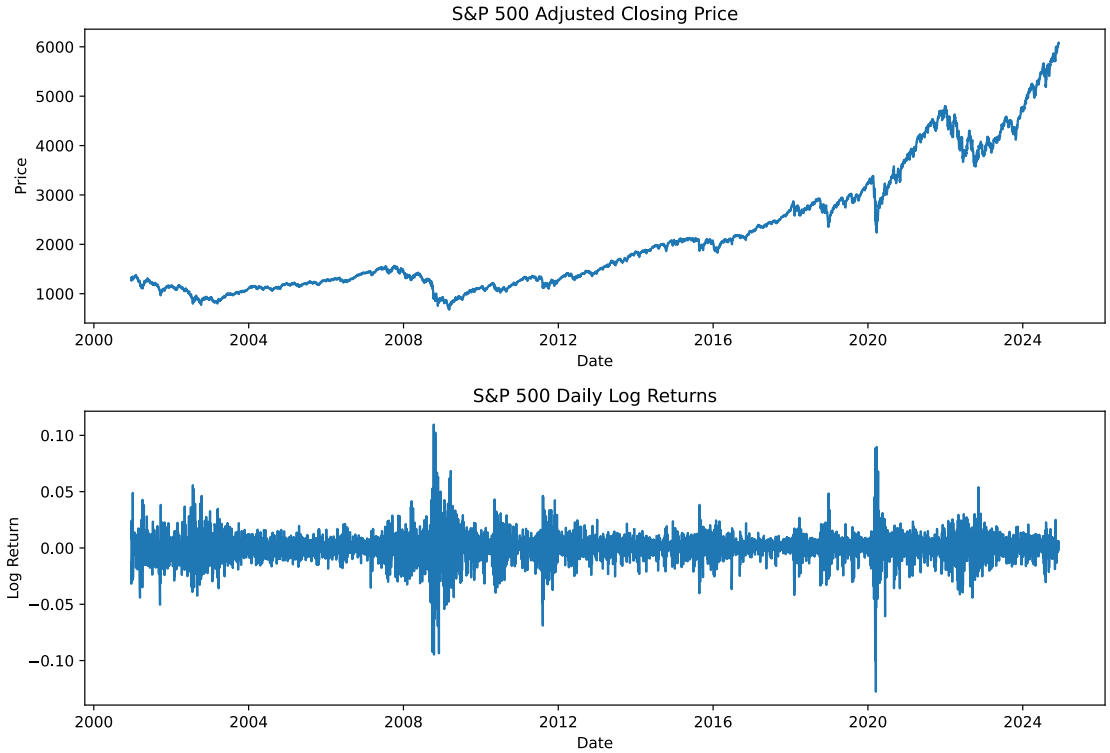


Figure: The adjusted closing price and the log daily returns of S&P 500 index.

Distributional Properties of SPX Returns

Leptokurtic and heavy-tailed.

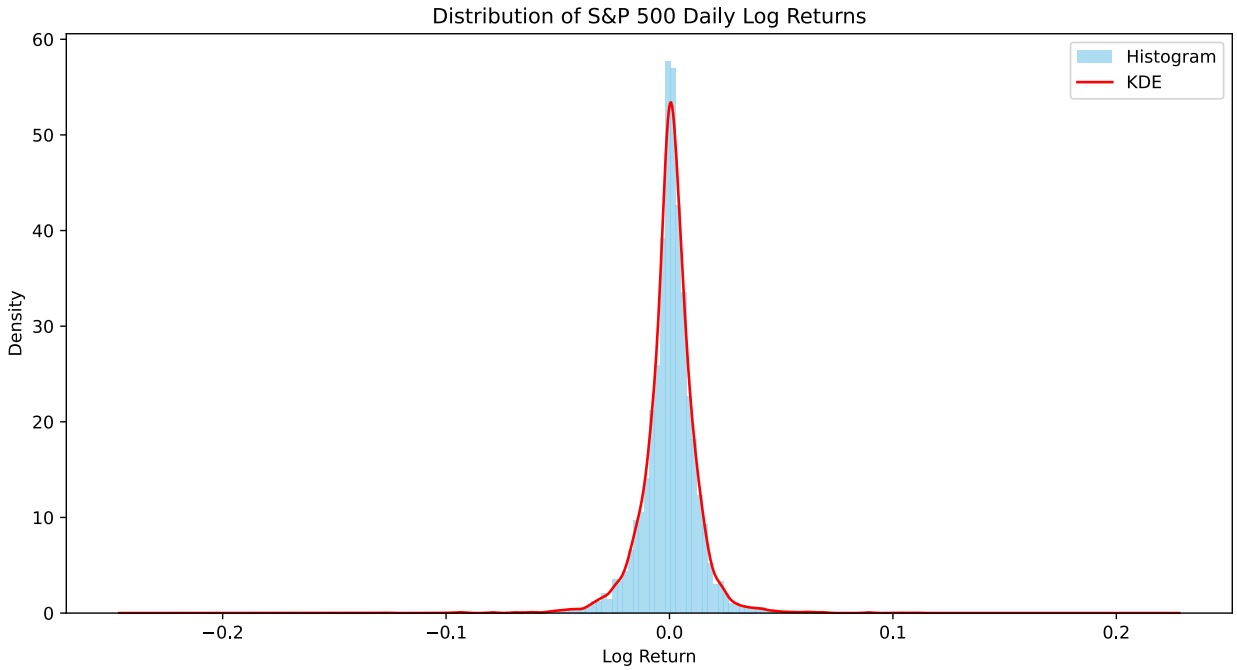


Figure: Distribution of S&P 500 Daily Log Returns.

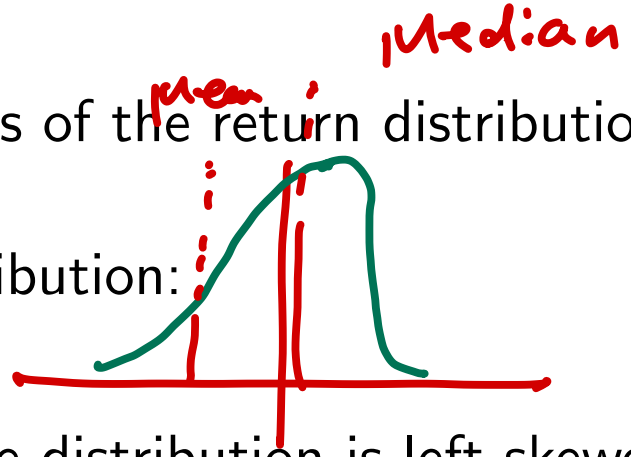
Some Key Statistics of SPX Returns

Statistic	**Value**
Mean	0.0003
Median	0.0007
Standard Deviation	0.0122
Skewness	−0.4077
Kurtosis	13.9822
Minimum	−0.1277
Maximum	0.1096

Table: Key Statistics of S&P 500 Daily Log Returns.

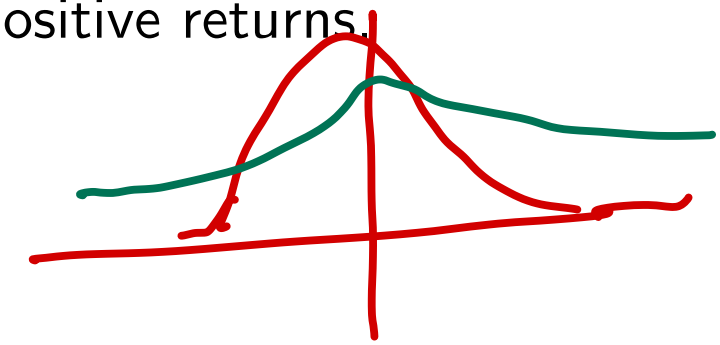
Skewness and Kurtosis

The statistics above reveal important properties of the return distribution.



Skewness measures the asymmetry of the distribution:

- Normal distribution has skewness of 0.
- Negative skewness (-0.4077) indicates the distribution is left-skewed.
- More extreme negative returns than positive returns.



Kurtosis measures the heaviness of tails:

- Normal distribution has kurtosis of 3.
- Much higher kurtosis (13.9822) indicates heavy tails.
- Heavy tails mean extreme returns occur more frequently than predicted by normal distribution.

Outline

1 Introduction to Univariate Data Analysis

2 Random Variables and Distributions

3 Estimation of Risk Measures

4 Extraction of Risk-Neutral Distributions from Option Data

Random Variables

A random variable is a mapping from the sample space to the real numbers.

$X : \Omega \to \mathbb{R}$.

- Ω is the sample space containing all possible outcomes of an experiment.
- $X(\omega)$ is the value of the random variable X at the outcome ω.

Example:

- $\Omega = \{HH, HT, TH, TT\}$: the sample space of the coin toss experiment.
- $X(\omega) =$ number of heads corresponding to $\omega \in \{HH, HT, TH, TT\}$.

Discrete Random Variables

Probability mass function (pmf):

$$p_X(x) = \mathbb{P}(X = x) = \mathbb{P}(\{\omega \in \Omega : X(\omega) = x\}).$$

Cumulative distribution function (cdf):

$$F_X(x) = \mathbb{P}(X \leq x) = \sum_{x_i \leq x} p_X(x_i).$$

Expectation of a discrete random variable:

$$\mathbb{E}[X] = \sum_x x p_X(x).$$

Continuous Random Variables

Cumulative distribution function (cdf):

$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt.$$

Probability density function (pdf):

$$f_X(x) = \frac{d}{dx}F_X(x).$$

Expectation of a continuous random variable:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)dx.$$

Moments of a Random Variable

The *k*-th moment of a random variable X is defined as:

$$\mathbb{E}[X^k] = \int_{-\infty}^{\infty} x^k f_X(x) dx.$$

The *k*-th central moment of a random variable X is defined as:

$$\mathbb{E}[(X - \mathbb{E}[X])^k] = \int_{-\infty}^{\infty} (x - \mathbb{E}[X])^k f_X(x) dx.$$

The moment generating function (mgf) of a random variable X is defined as:

$$M_X(t) = \mathbb{E}[e^{tX}].$$

$M_X^{(k)}(t) = E[(e^{tX})^{(k)}]$

$= E[X^k \cdot e^{tX}]$

We have that $\mathbb{E}[X^k] = M_X^{(k)}(0)$ (the k-th derivative of $M_X(t)$ at $t = 0$).

Skewness and Kurtosis of a Random Variable

Skewness and kurtosis are the third and fourth central normalized moments of a random variable, respectively.

The skewness of a random variable X is defined as:

$$\text{Skewness}(X) = \frac{\mathbb{E}[(X - \mathbb{E}[X])^3]}{(\mathbb{E}[(X - \mathbb{E}[X])^2])^{3/2}}.$$

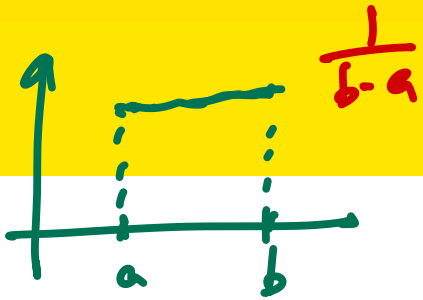
The kurtosis of a random variable X is defined as:

$$\text{Kurtosis}(X) = \frac{\mathbb{E}[(X - \mathbb{E}[X])^4]}{(\mathbb{E}[(X - \mathbb{E}[X])^2])^2}.$$

The excess kurtosis of a random variable X is defined as:

$$\text{Excess Kurtosis}(X) = \text{Kurtosis}(X) - 3.$$

Uniform Distribution



Property	**Formula/Value**
Parameters	$a < b$, $a, b \in \mathbb{R}$ (lower and upper bounds)
PDF	$f_X(x) = \frac{1}{b-a}$ for $x \in [a, b]$, 0 otherwise
MGF	$\frac{e^{bt}-e^{at}}{t(b-a)}$ for $t \neq 0$ and 1 for $t = 0$
Mean	$\mathbb{E}[X] = \frac{a+b}{2}$
Variance	$\mathrm{Var}(X) = \frac{(b-a)^2}{12}$
Skewness	0
Kurtosis	9/5

Normal Distribution

Property	**Formula/Value**
Parameters	$\mu \in \mathbb{R}$ (mean), $\sigma^2 > 0$ (variance)
PDF	$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$
MGF	$M_X(t) = \exp(\mu t + \frac{\sigma^2 t^2}{2})$
Mean	$\mathbb{E}[X] = \mu$
Variance	$\mathrm{Var}(X) = \sigma^2$
Skewness	0
Kurtosis	3

Table: Properties of Normal Distribution $X \sim N(\mu, \sigma^2)$.

Student's t Distribution

Property	**Formula/Value**
Parameters	$\nu > 0$ (degrees of freedom), $\mu \in \mathbb{R}$ (location), $\sigma^2 > 0$ (scale)
PDF	$f_X(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}\sigma}\left(1 + \frac{(x-\mu)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}}$
MGF	Does not exist for $t \neq 0$
Mean	$\mathbb{E}[X] = \mu$ for $\nu > 1$, otherwise undefined
Variance	$\mathrm{Var}(X) = \frac{\nu}{\nu-2}\sigma^2$ for $\nu > 2$, otherwise undefined
Skewness	0
Kurtosis	$3 + \frac{6}{\nu-4}$ for $\nu > 4$, otherwise undefined

The Student's t distribution converges to the normal distribution as the degrees of freedom increases to infinity.

Log-Normal Distribution

$Y = e^{\mu + \sigma Z}$, $Z \sim N(0,1)$

Property	**Formula/Value**
Parameters	$\mu \in \mathbb{R}$ (location), $\sigma^2 > 0$ (scale)
PDF	$f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right)$
MGF	Does not exist
Mean	$\mathbb{E}[X] = e^{\mu + \frac{\sigma^2}{2}}$
Variance	$\mathrm{Var}(X) = e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
Skewness	$(e^{\sigma^2} + 2)\sqrt{e^{\sigma^2} - 1}$
Kurtosis	$e^{4\sigma^2} + 2e^{3\sigma^2} + 3e^{2\sigma^2} - 3$

$E[e^{\mu+\sigma Z}] = e^{\mu} \cdot E[e^{\sigma Z}]$

$E[e^{2\mu+2\sigma Z}]$

> 3

Exponential Distribution

$\frac{1}{\lambda} e^{-\frac{x}{\lambda}}$

Property[1]	**Formula/Value**
Parameters	$\lambda > 0$ (rate)
PDF	$f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, 0 otherwise
MGF	$\frac{\lambda}{\lambda - t}$ for $t < \lambda$, otherwise undefined
Mean	$\mathbb{E}[X] = \frac{1}{\lambda}$
Variance	$\text{Var}(X) = \frac{1}{\lambda^2}$
Skewness	2
Kurtosis	9

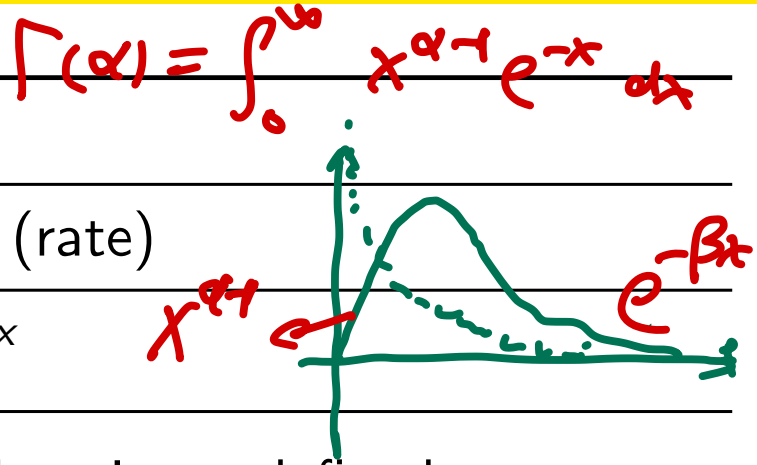
[1] Exponential distribution is often used to model the distribution of first arrival time.

Poisson Distribution

Property[2]	**Formula/Value**
Parameters	$\lambda > 0$ (rate)
PMF	$f_X(x) = \frac{\lambda^x}{x!} e^{-\lambda}, x \in \{0, 1, 2, \ldots\}$
MGF	$e^{\lambda(e^t - 1)}$
Mean	$\mathbb{E}[X] = \lambda$
Variance	$\mathrm{Var}(X) = \lambda$
Skewness	$\frac{1}{\sqrt{\lambda}}$
Kurtosis	$3 + \frac{1}{\lambda}$

[2] Poisson distribution is the distribution of the number of times an event occurs in a fixed interval of time or space, given the average number of times the event occurs in that interval.

Gamma Distribution



Property[3]	**Formula/Value**
Parameters	$\alpha > 0$ (shape), $\beta > 0$ (rate)
PDF	$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$
MGF	$\left(\frac{\beta}{\beta - t}\right)^\alpha$ for $t < \beta$, otherwise undefined
Mean	$\mathbb{E}[X] = \frac{\alpha}{\beta}$
Variance	$\mathrm{Var}(X) = \frac{\alpha}{\beta^2}$
Skewness	$\frac{2}{\sqrt{\alpha}}$
Kurtosis	$3 + \frac{6}{\alpha}$

[3] Gamma distribution is a continuous probability distribution that is often used to model the time until an event occurs in a process that follows an exponential distribution.

Inverse Gaussian Distribution

Property	**Formula/Value**
Parameters	$\mu \in \mathbb{R},\ \lambda > 0$
PDF	$f_X(x) = \sqrt{\frac{\lambda}{2\pi x^3}} \exp\left(-\frac{\lambda(x-\mu)^2}{2\mu^2 x}\right),\ x > 0$
MGF	$\exp\left(\frac{\lambda}{\mu}\left(1 - \sqrt{1 - 2\mu^2 t/\lambda}\right)\right)$
Mean	$\mathbb{E}[X] = \mu$
Variance	$\mathrm{Var}(X) = \mu^3/\lambda$
Skewness	$3\sqrt{\mu/\lambda}$
Kurtosis	$3 + 15\mu/\lambda$

Normal Inverse Gaussian Distribution

Property[4]	**Formula/Value**
Parameters	$\mu \in \mathbb{R}$ (location), $\alpha > 0$ (tail heaviness), $\beta \in \mathbb{R}$ (asymmetry), $\delta > 0$ (scale), $\gamma = \sqrt{\alpha^2 - \beta^2}$
PDF	$f_X(x) = \frac{\alpha\delta}{\pi\sqrt{\delta^2+(x-\mu)^2}} K_1\left(\alpha\sqrt{\delta^2+(x-\mu)^2}\right) \exp(\delta\gamma + \beta(x-\mu))$
MGF	$\exp(\mu t + \delta(\gamma - \sqrt{\alpha^2 - (\beta+t)^2}))$ for $\lvert\beta+t\rvert < \alpha$, otherwise undefined
Mean	$\mathbb{E}[X] = \mu + \delta\beta/\gamma$
Variance	$\mathrm{Var}(X) = \delta\alpha^2/\gamma^3$
Skewness	$3\beta/(\alpha\sqrt{\delta\gamma})$
Kurtosis	$3 + 3(1 + 4\beta^2/\alpha^2)/(\delta\gamma)$

[4] Normal Inverse Gaussian distribution is a generalization of the Normal distribution that allows for both location and scale parameters.

[5] $K_1(x)$ is the modified Bessel function of the second kind of order 1.

Chi-Square Distribution

Property[6]	**Formula/Value**
Parameters	$k > 0$ (degrees of freedom)
PDF	$f_X(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}$
MGF	$(1-2t)^{-k/2}$ for $t < 1/2$, otherwise undefined
Mean	$\mathbb{E}[X] = k$
Variance	$\mathrm{Var}(X) = 2k$
Skewness	$\sqrt{8/k}$
Kurtosis	$3 + \frac{12}{k}$

[6] Chi-square distribution is the distribution of the sum of the squares of k independent standard normal random variables.

F Distribution

Property[7]	**Formula/Value**
Parameters	$d_1 > 0$, $d_2 > 0$ (degrees of freedom)
PDF	$f_X(x) = \frac{\sqrt{(d_1 x)^{d_1} d_2^{d_2}}}{x B(d_1/2, d_2/2)} \frac{1}{(d_1 x + d_2)^{(d_1+d_2)/2}}$
MGF	Does not exist
Mean	$\mathbb{E}[X] = \frac{d_2}{d_2 - 2}$ for $d_2 > 2$, otherwise undefined
Variance	$\mathrm{Var}(X) = \frac{2d_2^2(d_1+d_2-2)}{d_1(d_2-2)^2(d_2-4)}$ for $d_2 > 4$, otherwise undefined
Skewness	$\frac{2(2d_1+d_2-2)\sqrt{8(d_2-4)}}{(d_2-6)\sqrt{d_1(d_1+d_2-2)}}$ for $d_2 > 6$, otherwise undefined
Kurtosis	$3 + \frac{12(d_1(5d_2-22)+(d_2-4)^2)}{d_1(d_2-6)(d_2-8)}$ for $d_2 > 8$, otherwise undefined

[7] F distribution is the distribution of the ratio of two chi-square random variables.

Beta Distribution

Property[8]	**Formula/Value**
Parameters	$\alpha > 0$ (shape), $\beta > 0$ (shape)
PDF	$f_X(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$
MGF	$\frac{1}{B(\alpha,\beta)} \int_0^1 e^{tx} x^{\alpha-1}(1-x)^{\beta-1} dx$
Mean	$\mathbb{E}[X] = \frac{\alpha}{\alpha+\beta}$
Variance	$\mathrm{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Skewness	$\frac{2(\beta-\alpha)\sqrt{\alpha+\beta+1}}{(\alpha+\beta+2)\sqrt{\alpha\beta}}$
Kurtosis	$\frac{6((\alpha-\beta)^2(\alpha+\beta+1)-\alpha\beta(\alpha+\beta+2))}{(\alpha\beta(\alpha+\beta+2)(\alpha+\beta+3))} + 3$

[8] Beta distribution is is often used to model the distribution of a random variable that takes values in a finite interval [0, 1].

Fitting SPX Returns by Normal Distribution

$r_t = \ln \frac{S_t}{S_{t-1}}$: $r_1, r_2, \ldots, r_T$

A normal distribution is fully characterized by its mean and variance, we take the mean and variance of SPX returns as the parameters of the normal distribution (method of moments):

$$\widehat{\mu} = \mathbb{E}[X] = \frac{1}{T}\sum_{t=1}^{T} r_t,$$

$r_t \sim$ i.i.d $N(\mu, \sigma^2)$

$E[\hat{\sigma}^2] = \sigma^2$

$$\widehat{\sigma}^2 = \mathrm{Var}(X) = \frac{1}{T-1}\sum_{t=1}^{T}(r_t - \widehat{\mu})^2.$$

The denominator of the variance is $T-1$ instead of T because we use the sample mean to estimate the population mean.

The Method of Moments

The method of moments is a simple and intuitive method for estimating the parameters of a distribution.

$$\hat{M}_1 = M_1(\Theta)$$
$$\hat{M}_2 = M_2(\Theta)$$
$$\vdots$$
$$\hat{M}_k = M_k(\Theta)$$

To match the first k moments of the distribution, we set the first k sample moments equal to the first k population moments.

The parameters are then the solutions to the equations.

The method of moments is simple and intuitive, but it has several drawbacks:

- The estimate parameters may be outside of the parameter space (in frequent for small samples);
- The estimates by the method of moments are not necessarily sufficient statistics, i.e., they sometimes fail to take into account all relevant information in the sample.

The Maximum Likelihood Estimation

The maximum likelihood estimation (MLE) is a method for estimating the parameters of a distribution by maximizing the likelihood function.

$p(x;\theta)$: $x_1, x_2 \ldots, x_n$

The log likelihood function is defined as:

$L(\theta) = p(x_1;\theta)\, p(x_2;\theta) \cdots p(x_n;\theta)$

$$\max \quad \log L(\theta) = \sum_{i=1}^{n} \log p(x_i;\theta)$$

where $p(x_i;\theta)$ is the probability density function of the distribution, and θ is the parameter vector.

The MLE is the parameter vector that maximizes the log ikelihood function.

For a normal ditribution, the MLE estimator is the same as the method of moments estimator.

$$p(r_t; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(r_t-\mu)^2}{2\sigma^2}}$$

$$\sum \log p = \sum_t -\frac{(r_t-\mu)^2}{2\sigma^2} - \log \frac{1}{\sqrt{2\pi\sigma^2}}$$

Goodness of Fit of SPX Returns by Normal Distribution

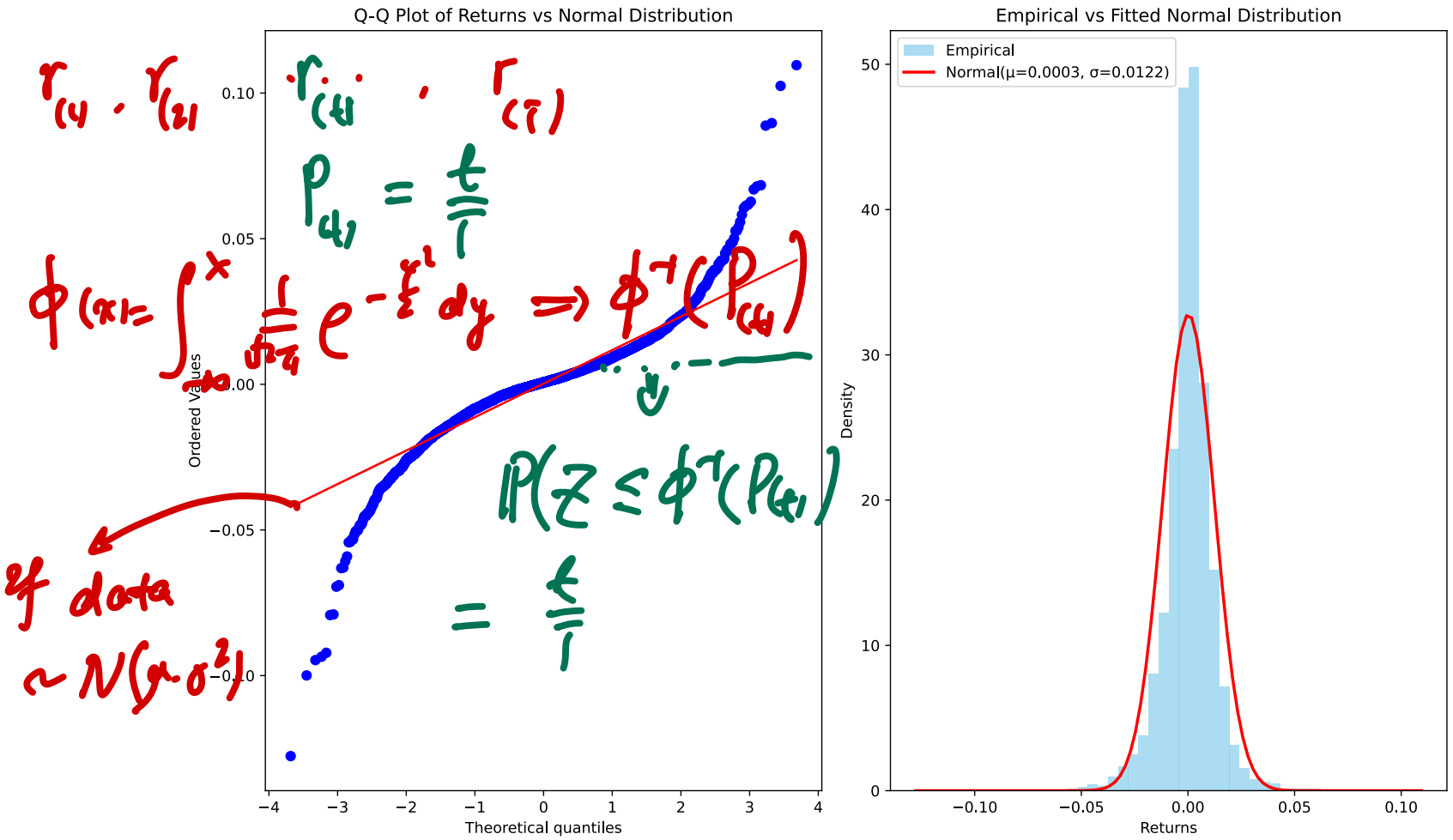


Figure: Goodness of Fit of SPX Returns by Normal Distribution.

Q-Q Plot

The Q-Q plot is a graphical method for assessing the fit of a distribution to a sample.

It is a plot of the quantiles of the sample against the quantiles of the distribution.

If the sample is drawn from the distribution, the points in the Q-Q plot will lie on the line $y = x$.

From the Q-Q plot, we can see that the SPX returns do not fit the normal distribution well.

Hypothesis Testing

The hypothesis testing is a statistical method for testing whether a hypothesis is true or false.

The null hypothesis is the hypothesis that we want to test, and the alternative hypothesis is the hypothesis that we want to reject.

The test statistic is a function of the sample data that is used to test the null hypothesis.

Small p: reject .. — 0.05
large p: fail to reject 0.01
0.1

The p-value is the probability of making a type I error, i.e., rejecting the null hypothesis when it is true.

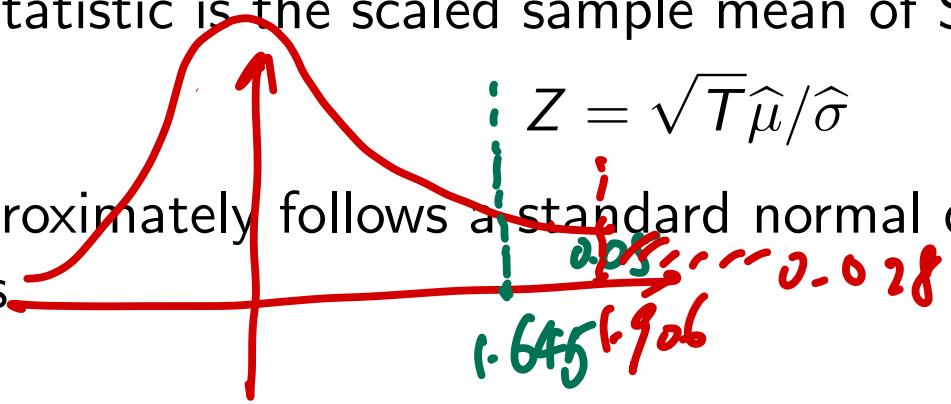
To calculate the p-value, we need to know the distribution of the test statistic under the null hypothesis.

Example: Test Whether the Mean of SPX Returns is Greater Than Zero

The null hypothesis is that the mean of SPX returns is zero, and the alternative hypothesis is that the mean of SPX returns is greater than zero.

The test statistic is the scaled sample mean of SPX returns:

$$Z = \sqrt{T}\hat{\mu}/\hat{\sigma}$$

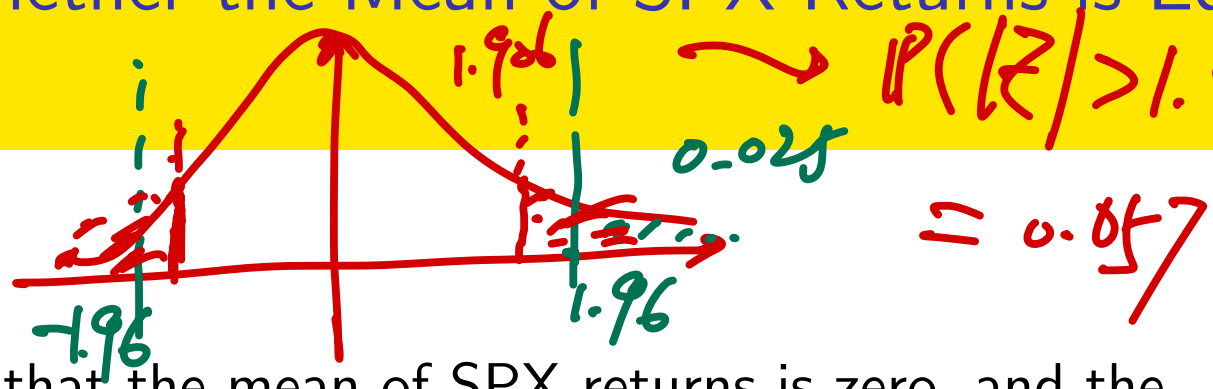


which approximately follows a standard normal distribution under the null hypothesis.

The p-value is the probability (0.028) of the test statistic being greater than the observed ($\sqrt{6028} \times 0.0003/0.0122 = 1.906$) under the null hypothesis.

The threshold of the statistic for rejecting the null hypothesis at the 5% significance level is 1.645.

Example: Test Whether the Mean of SPX Returns is Equal to Zero



The null hypothesis is that the mean of SPX returns is zero, and the alternative hypothesis is that the mean of SPX returns is not zero.

The p-value is the probability (0.057) of the test statistic being greater than or equal to the observed ($\sqrt{6028} \times 0.0003/0.0122 = 1.906$) in absolute value under the null hypothesis.

The threshold of the statistic for rejecting the null hypothesis at the 5% significance level is 1.96.

The Shapiro-Wilk Test

The Shapiro-Wilk test is a test of normality.

The test statistic is given by:

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

where $x_{(i)}$ is the i-th order statistic of the sample, and a_i is the i-th element of the vector $(a_1, a_2, \ldots, a_n)$ given by:

$$a_i = \frac{m_i}{\sqrt{m_0 m_i}},$$

where m_i is the i-th moment of the standard normal distribution, and m_0 is the sample mean.

The Shapiro-Wilk Test (cont'd)

Under the normal distribution, the test statistic W follows the distribution:

$$W \sim \text{Beta}\left(\frac{n}{2}, \frac{n}{2}\right)$$

where n is the sample size.

The p-value is the probability of making a type I error, i.e., rejecting the null hypothesis when it is true:

$$p = P(W \leq w_{\text{obs}} | H_0) = \int_0^{w_{\text{obs}}} f_W(w) dw$$

where w_{obs} is the observed value of the test statistic and $f_W(w)$ is the probability density function of the Beta$(\frac{n}{2}, \frac{n}{2})$distribution.

Testing Normality of SPX Returns

Apart from visual inspection, we can also use hypothesis testing to test whether the sample is drawn from the distribution.

The null hypothesis is that the sample is drawn from the distribution, and the alternative hypothesis is that the sample is not drawn from the distribution.

We can use the Shapiro-Wilk test to test whether the sample is drawn from the normal distribution.

The p-value of the Shapiro-Wilk test is 0.0000, which is less than 0.05. Therefore, we reject the null hypothesis and conclude that the SPX returns do not follow the normal distribution.

Fitting SPX Returns by Student's t Distribution

The Student's t distribution is a generalization of the normal distribution that allows for heavy tails.

We fit the Student's t distribution to the SPX returns by maximum likelihood estimation (MLE):

$$\max_{\nu,\mu,\sigma} \log L(\nu, \mu, \sigma) = \sum_{t=1}^{T} \log p(r_t; \nu, \mu, \sigma).$$

Goodness of Fit of SPX Returns by Student's t Distribution

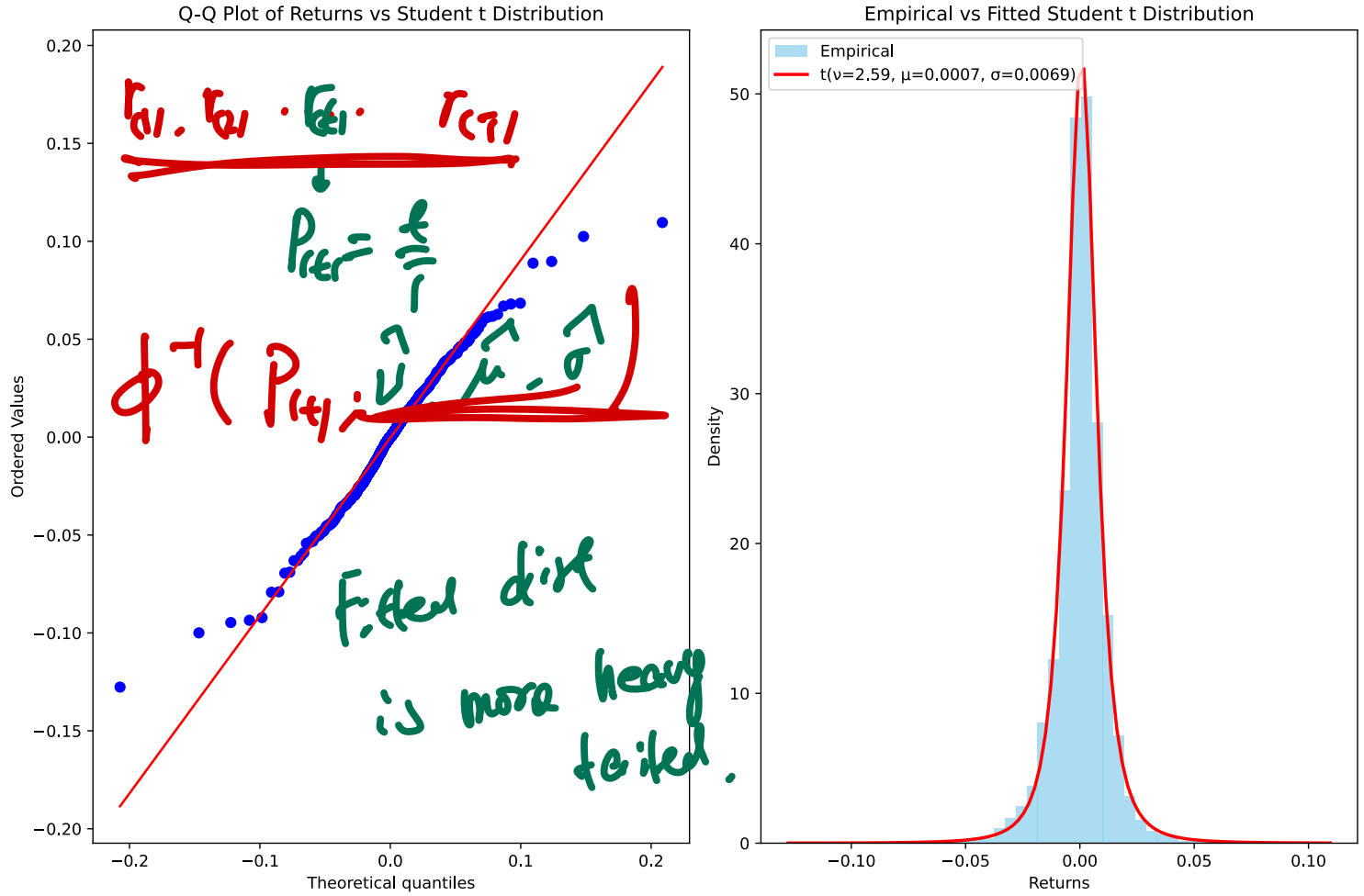


Figure: Goodness of Fit of SPX Returns by Student's t Distribution.

The Kolmogorov-Smirnov Test

The Kolmogorov-Smirnov test is to test whether the sample is drawn from the distribution.

$F_n(x)$ = Proportion of data $\leq x$

The test statistic is given by:

$$D_n = \sup_{x \in \mathbb{R}} |F_n(x) - F(x)|$$

where $F_n(x)$ is the empirical distribution function of the sample, and $F(x)$ is the distribution function of the hypothesized distribution.

Under the null hypothesis, the test statistic $\sqrt{n}D_n$ converges to the Kolmogorov distribution:

$$P(\sqrt{n}D_n > c) \approx 2 \sum_{k=1}^{\infty} (-1)^{k-1} e^{-2k^2c^2}.$$

Test the Fit of SPX Returns by Student's t Distribution

We conduct the Kolmogorov-Smirnov test to test whether the SPX returns follow the Student's t distribution.

0.0318

The p-value of the Kolmogorov-Smirnov test is 3.18e-02, which is less than 0.05. Therefore, we reject the null hypothesis and conclude that the SPX returns do not follow the Student's t distribution.

Normal: $e^{-\frac{x^2}{2}}$ $\Rightarrow$ $e^{-\alpha x}$

t : $x^{-|\alpha+1|}$

Fitting SPX Returns by the Normal Inverse Gaussian Distribution

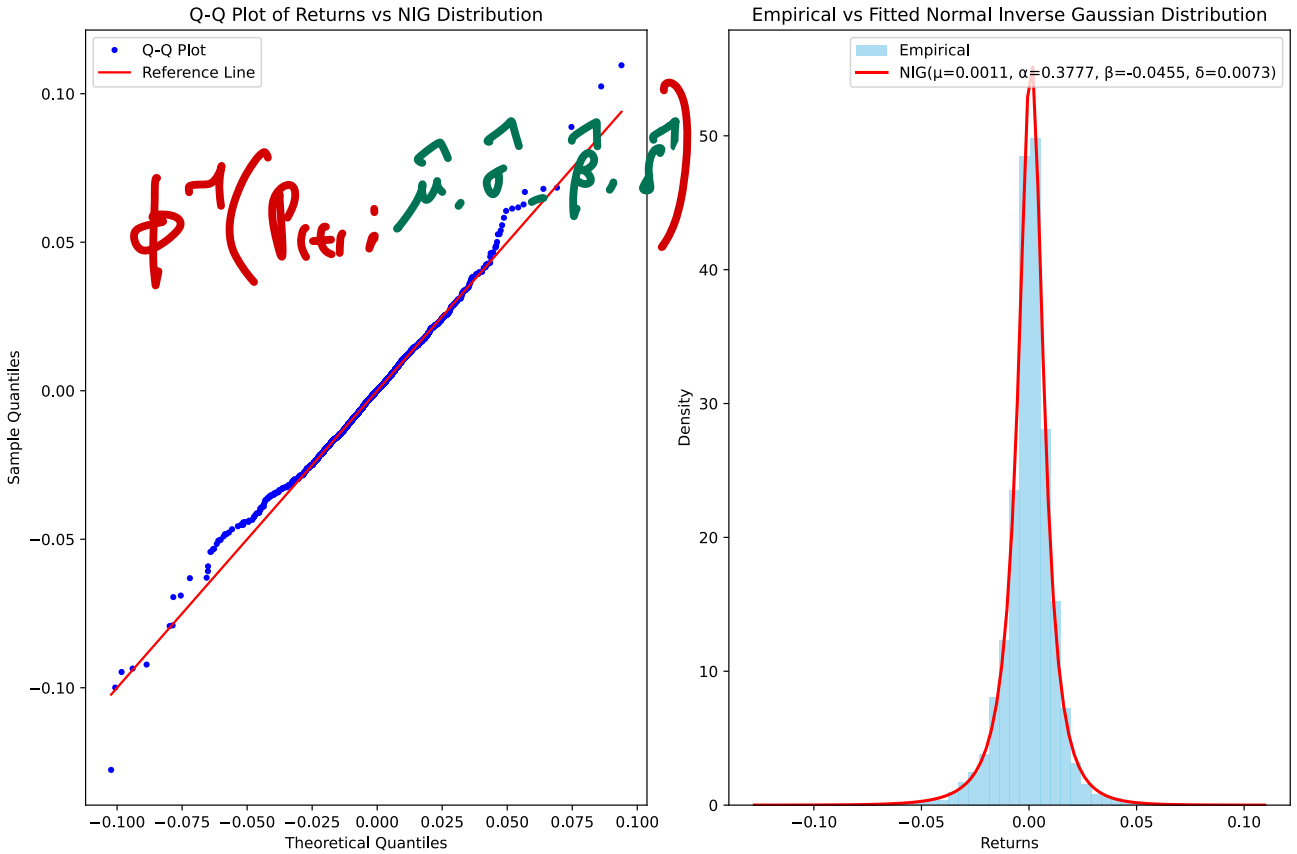


Figure: Goodness of Fit of SPX Returns by the Normal Inverse Gaussian Distribution.

Kolmogorov-Smirnov Test for SPX Returns by the Normal Inverse Gaussian Distribution

Kolmogorov-Smirnov Test for Normal Inverse Gaussian Distribution:

Measure	**Value**
Statistic	0.0117
p-value	0.3792

Since the p-value (0.3792) is much larger than the significance level of 0.05, we fail to reject the null hypothesis.

This suggests that the SPX returns can be well approximated by the Normal Inverse Gaussian distribution.

Outline

1 Introduction to Univariate Data Analysis

2 Random Variables and Distributions

3 Estimation of Risk Measures

4 Extraction of Risk-Neutral Distributions from Option Data

Risk Measures

Risk measures are used to quantify the risk of a portfolio.

Risk measures are useful for risk management, performance evaluation, portfolio construction, regulatory compliance, etc.

Examples of applications:

- Evaluate risk-adjusted performance of different trading strategies;
- Determine margin requirements for short positions or derivatives;
- Determine the capital requirement for a bank or insurance company;
- As criteria for trading and investment decisions;
- ...

Risk Measures (cont'd)

Suppose we have a portfolio that has a return X in a future time horizon T.

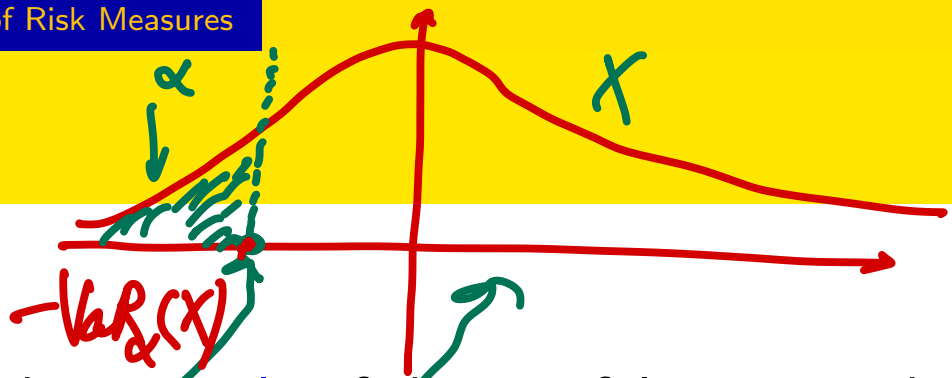
From the current perspective, the return X is uncertain, and but we would like to quantify the risk of X, i.e, to establish a mapping from X to a real number:

$$\rho : \mathcal{X} \to \mathbb{R}$$

where $\mathcal{X}$ is the space of random variables.

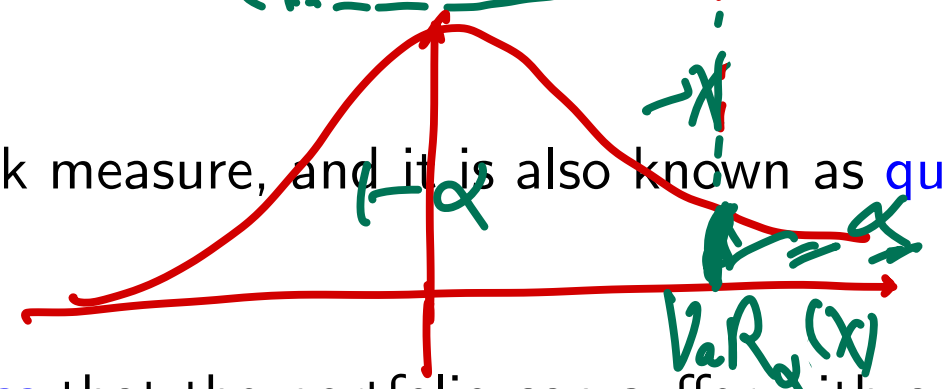
The function ρ is called a risk measure.

Value-at-Risk (VaR)



The value-at-risk (VaR) is the quantile of the portfolio return distribution:

$$\text{VaR}_\alpha(X) := -\inf\{x \in \mathbb{R} : \mathbb{P}(X \leq x) > \alpha\} = F_{-X}^{-1}(1-\alpha).$$



The VaR is a popular risk measure, and it is also known as quantile risk measure.

It gives the maximum loss that the portfolio can suffer with a confidence level of $1-\alpha$.

Typically, the ~~confidence level is~~ α set to be 1% or 5%.

Value-at-Risk (VaR)

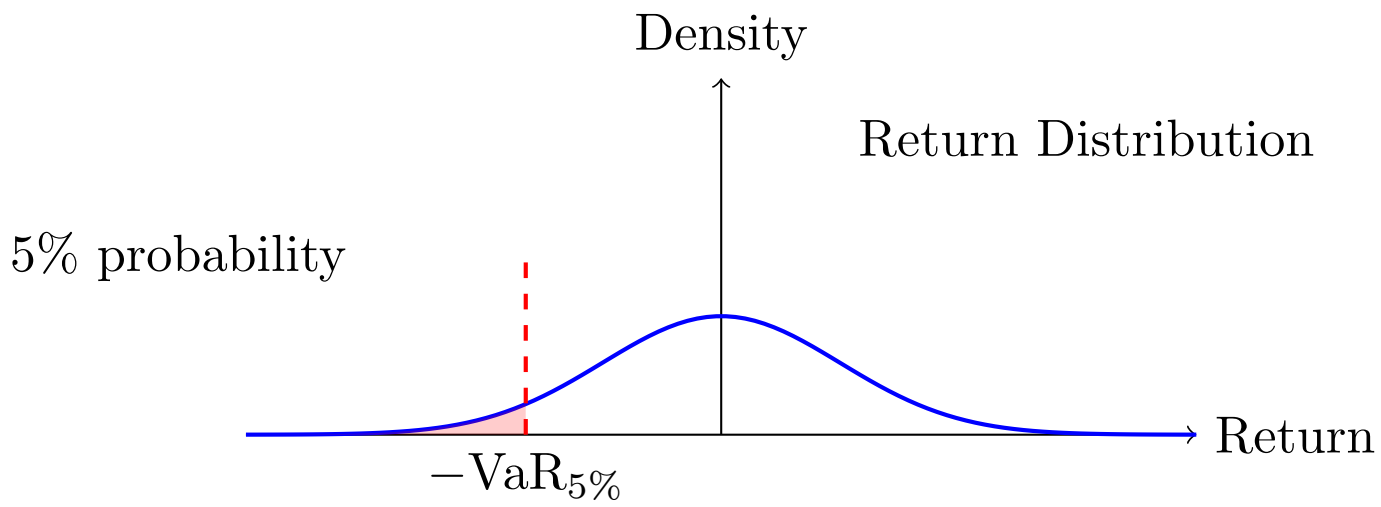


Figure: Illustration of VaR.

Conditional Value-at-Risk (CVaR)

The Conditional Value-at-Risk (CVaR), also known as Expected Shortfall (ES), is defined as:

$$\text{CVaR}_\alpha(X) := \mathbb{E}[-X | -X \geq \text{VaR}_\alpha(X)] = \frac{1}{\alpha} \int_0^\alpha \text{VaR}_\beta(X) d\beta.$$

CVaR measures the expected loss given that the loss exceeds VaR.

Properties of CVaR:

- CVaR is always greater than or equal to VaR.
- CVaR takes into account the tail behavior of the distribution.
- CVaR is a coherent risk measure.
- CVaR is more sensitive to extreme events than VaR.

Conditional Value-at-Risk (CVaR)

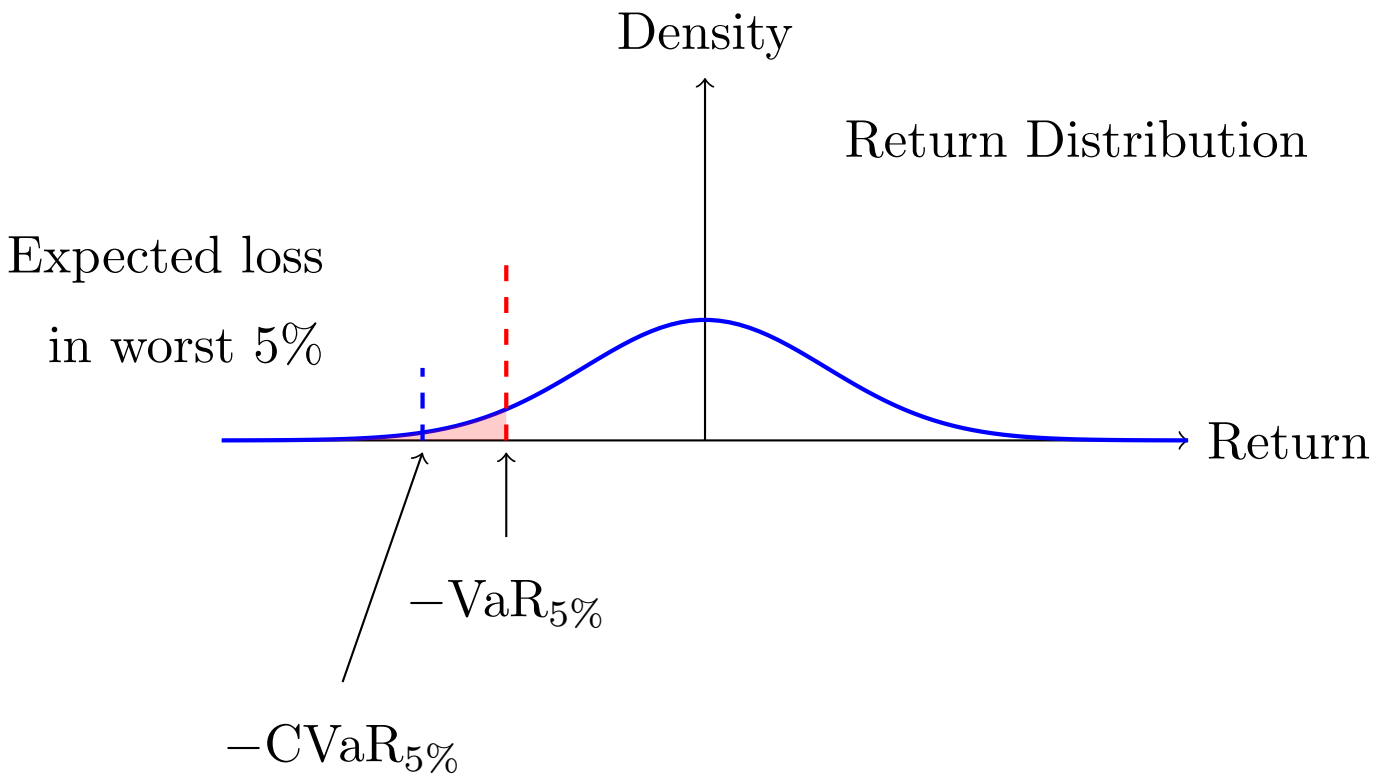


Figure: Illustration of CVaR.

Coherent Risk Measures

Coherent risk measures are a class of risk measures that satisfy the following properties:

- Normalized: $\rho(0) = 0$.
- Monotonicity: If $X \leq Y$, then $\rho(X) \geq \rho(Y)$.
- Subadditivity: $\rho(X + Y) \leq \rho(X) + \rho(Y)$.
- Positive homogeneity: $\rho(\lambda X) = \lambda \rho(X)$ for all $\lambda \geq 0$.
- Translation invariance: $\rho(X + A) = \rho(X) - a$ for all $A = a$ a.s.

The VaR is not a coherent risk measure as it does not satisfy the subadditivity property.

Historical Simulation for Estimation of Risk Measures

We may use the historical simulation to estimate the risk measures.

The historical simulation is a non-parametric method that uses the historical data to estimate the risk measures.

The historical simulation is a simple method that does not require any distributional assumptions.

The procedure of historical simulation is as follows:

- Simulate the portfolio returns from the historical data .
- Sort the simulated returns in ascending order.
- Estimate the risk measures using the simulated returns.

Historical Simulation for Estimation of VaR

Suppose we have simulated a sample of historical returns $\{X_1, \ldots, X_T\}$.

We can estimate the VaR at confidence level α as:

$$\text{VaR}_\alpha(X) = -X_{(\lceil T\alpha \rceil)}$$

where:

- $X_{(i)}$ is the *i*-th order statistic of the sample $\{X_1, \ldots, X_T\}$;
- $\lceil \cdot \rceil$ denotes the ceiling function that returns the smallest integer greater than or equal to the input.

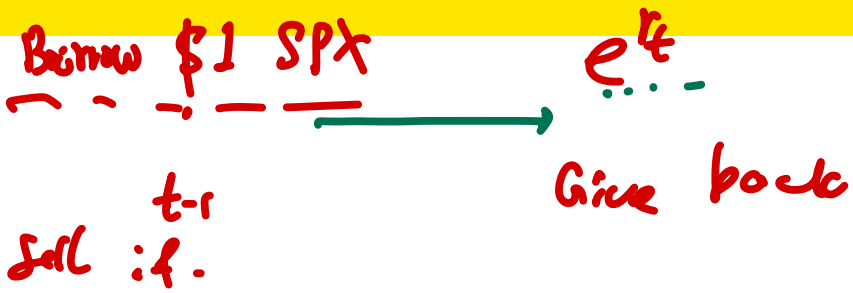
Historical Simulation for Estimation of CVaR

We can estimate the CVaR at confidence level α as:

$$\text{CVaR}_\alpha(X) = -\frac{1}{\lceil T\alpha \rceil} \sum_{i=1}^{\lceil T\alpha \rceil} X_{(i)},$$

which is the average of the largest $\lceil T\alpha \rceil$ losses.

Caculate VaR and CVaR of of Shorting 'SPX



As an example, we calculate the daily VaR and CVaR of shorting SPX.

1 $-e^{r_t}$

For this position, the daily return is the negative of the SPX return:

$$X_t = 1 - e^{r_t^{\text{SPX}}}.$$

We regard the return sample $\{X_1, \ldots, X_T\}$ as the historical simulation data, and use it to estimate the VaR and CVaR at confidence level α.

VaR and CVaR of Shorting SPX

Confidence Level	VaR	CVaR
1.00%	0.0343	0.0493
5.00%	0.0184	0.0293
10.00%	0.0124	0.0222

Table: Risk Measures for Shorting SPX Position at Different Confidence Levels.

VaR and CVaR of Shorting SPX: 10% Confidence Level

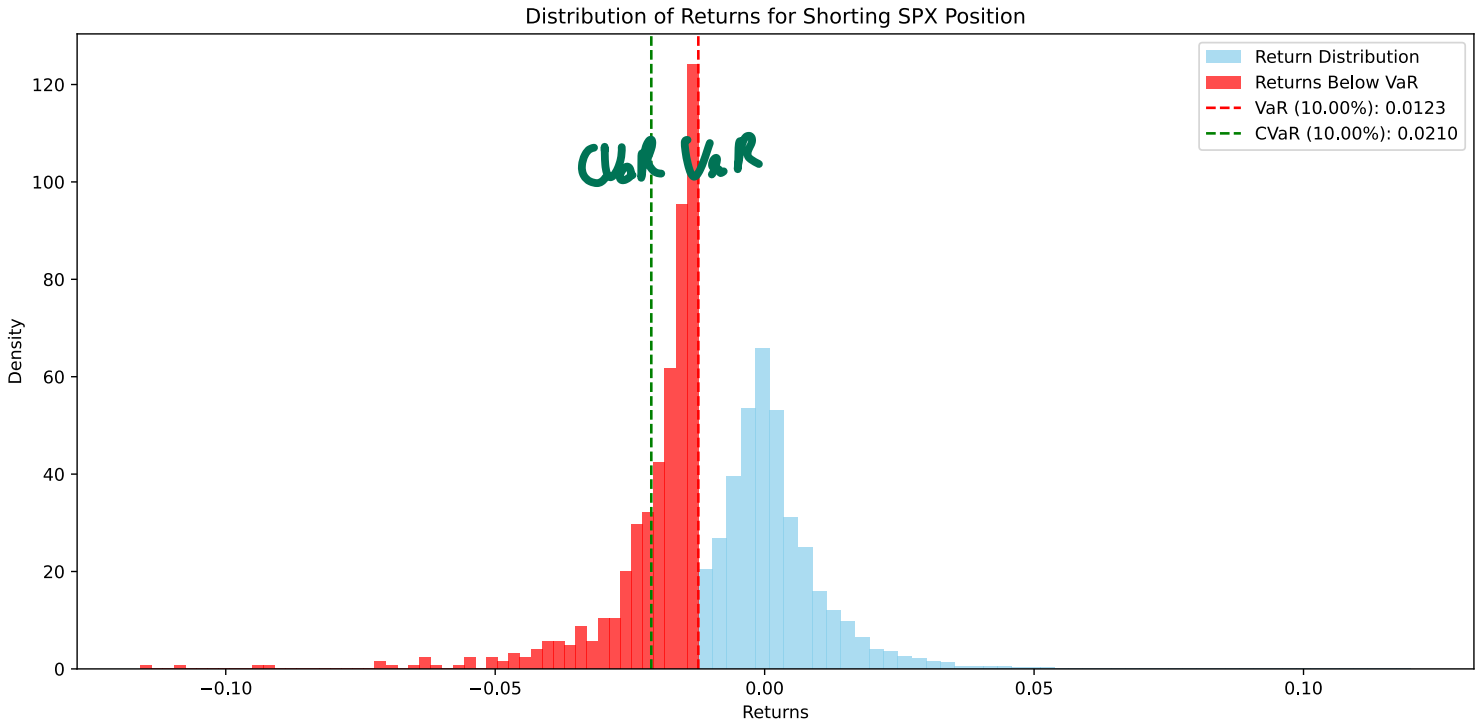


Figure: VaR and CVaR of Shorting SPX at 10% Confidence Level.

VaR and CVaR of Shorting SPX: 5% Confidence Level

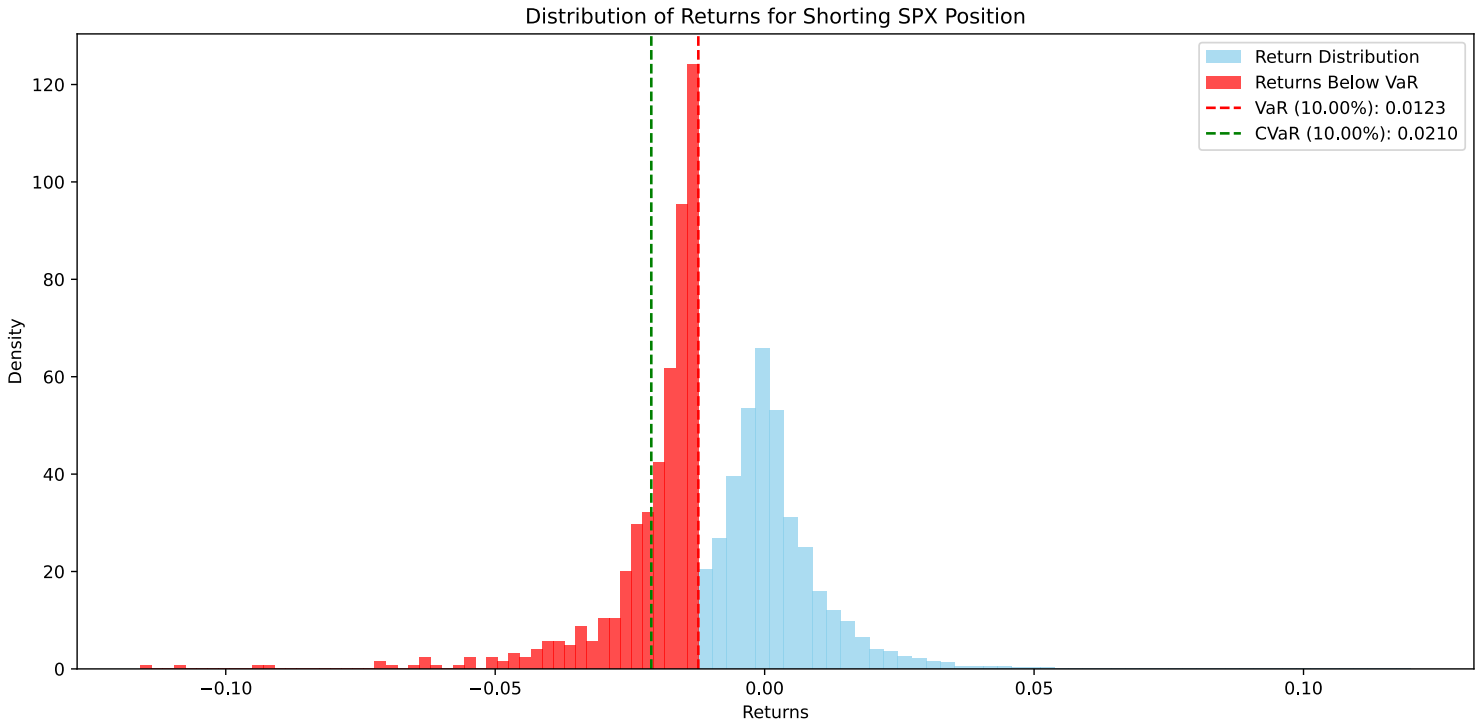


Figure: VaR and CVaR of Shorting SPX at 10% Confidence Level.

VaR and CVaR of Shorting SPX: 1% Confidence Level



Figure: VaR and CVaR of Shorting SPX at 1% Confidence Level.

The estimations tend to be unreliable when the confidence level is low as the sample size is small.

Estimation of Risk Measures by Parametric Models

Confidence Level	Method	VaR	CVaR
0.10%	Historical	0.0692	0.0887
0.10%	Normal	0.0386	0.0397
0.10%	Student-t	0.0946	0.1076
0.10%	NIG	0.0614	0.0656
0.50%	Historical	0.0425	0.0597
0.50%	Normal	0.0321	0.0352
0.50%	Student-t	0.0493	0.0689
0.50%	NIG	0.0413	0.0505
1.00%	Historical	0.0339	0.0488
1.00%	Normal	0.0290	0.0327
1.00%	Student-t	0.0371	0.0551
1.00%	NIG	0.0334	0.0434
5.00%	Historical	0.0169	0.0277
5.00%	Normal	0.0205	0.0255
5.00%	Student-t	0.0183	0.0307
5.00%	NIG	0.0178	0.0273

Table: Comparison of Risk Measures across Different Methods.

Comparison of Fitted Distributions of SPX Returns

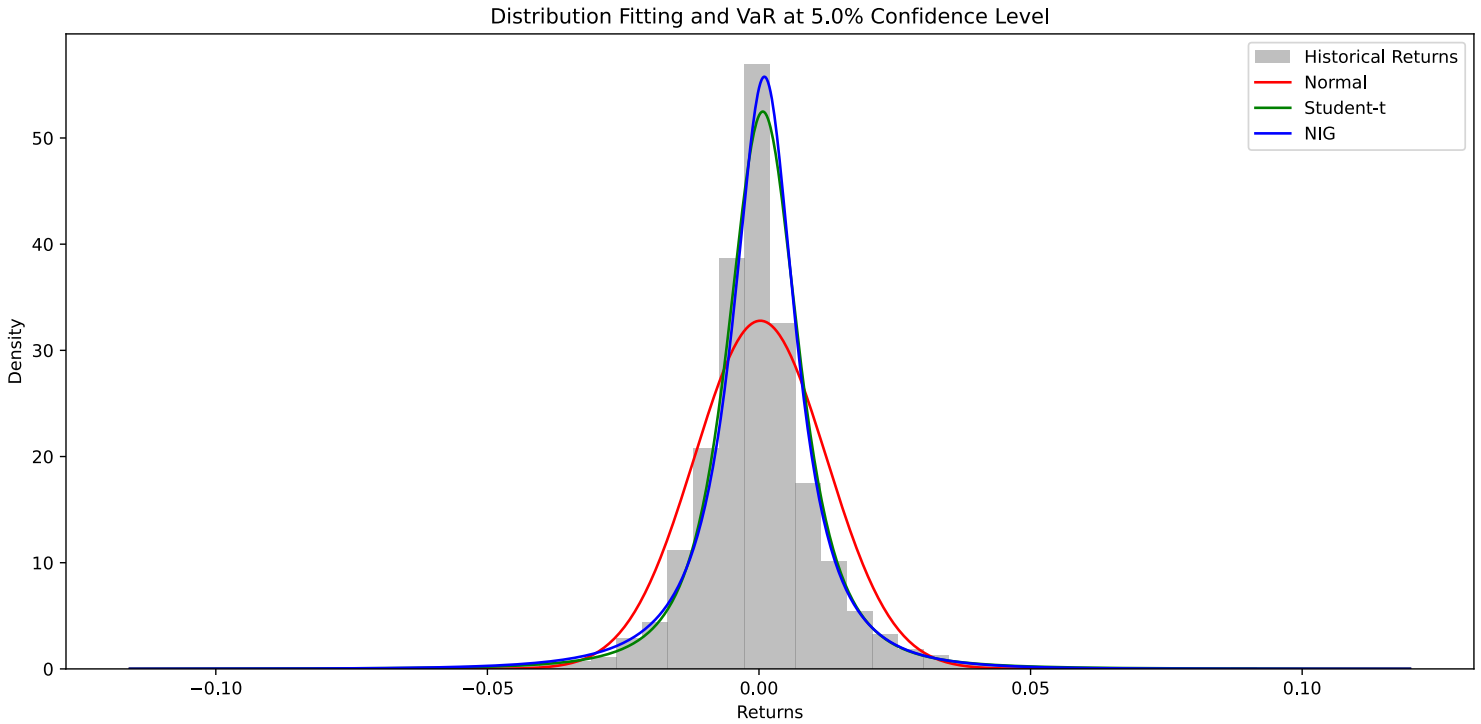


Figure: Comparison of Fitted Distributions of SPX Returns.

Comparison of Fitted Distributions of SPX Returns: Tail Region

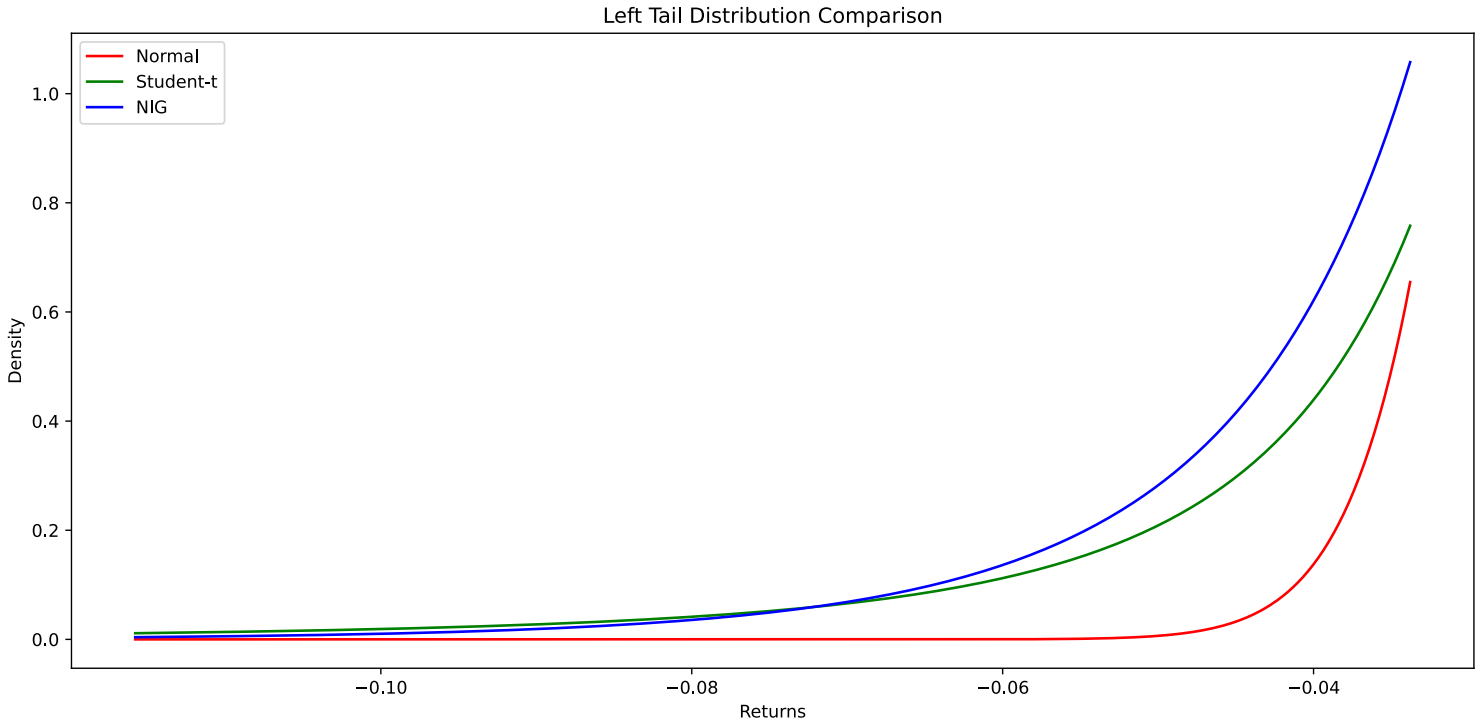


Figure: Comparison of Fitted Distributions of SPX Returns.

Outline

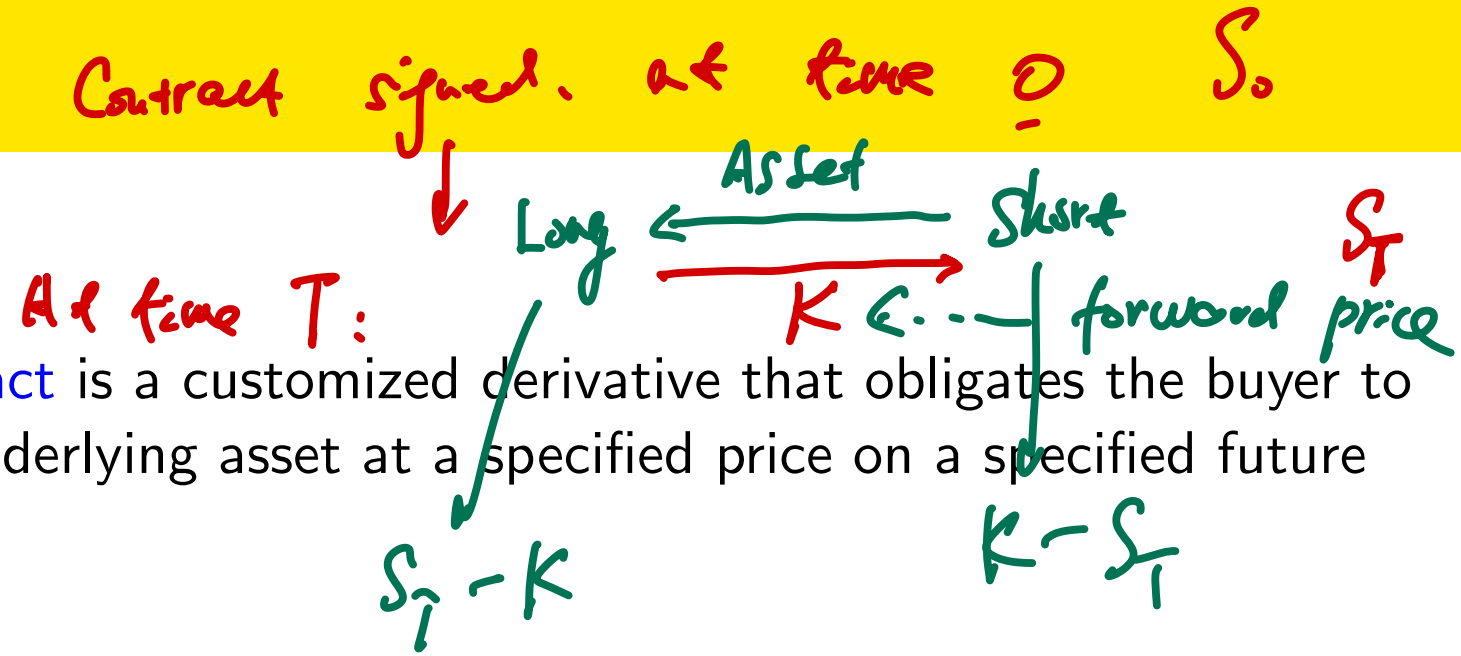
1 Introduction to Univariate Data Analysis

2 Random Variables and Distributions

3 Estimation of Risk Measures

4 Extraction of Risk-Neutral Distributions from Option Data

Forwards



A forward contract is a customized derivative that obligates the buyer to buy or sell an underlying asset at a specified price on a specified future date.

The specified price is called the forward price.

The payoff of a forward contract is given by:

$$\text{Payoff} = S_T - K,$$

where K is the forward price.

Options



Options are a type of financial derivatives that give the buyer the right, but not the obligation, to buy or sell an underlying asset at a specified price on or before a certain date.

The underlying asset can be a stock, an index, a bond, a commodity, or a currency.

The specified price is called the strike price or exercise price.

The specified date is called the expiration date or maturity date.

The buyer of an option pays a premium (option price) to the seller of the option.

European Call and Put Options

A European call option grants the buyer the right to buy the underlying asset at the strike price K at the maturity date.

$$\text{Payoff} = \max(S_T - K, 0).$$

$S_T > K$: $S_T - K$

$S_T < K$: not exercise

0

A European put option grants the buyer the right to sell the underlying asset at the strike price K at the maturity date.

$$\text{Payoff} = \max(K - S_T, 0).$$

$S_T > K$: not exercise

0

$S_T < K$: $K - S_T$

Other types of options include American options, which can be exercised at any time before the maturity date, and exotic options, which have payoffs that are more complex than the standard call and put options.

Risk-Neutral Pricing of Options

Option: $f(S_T)$

$\mathbb{E}^{\mathbb{P}}[e^{-rT} f(S_T)]$

Ex: $\max(K - S_T, 0)$

SPX index

Take effect of hedging

Generally speaking, the option price is determined by the distribution of the underlying asset price at the maturity date S_T, and market's risk preferences.

risk of holding SPX

$S_T < K$

Then the option price can be expressed as:

$M_T(S_T)$

$$C = \mathbb{E}_{\mathbb{P}}[M_T e^{-rT} f(S_T)],$$

where M_T a non-negative random variable representing risk preferences, and $f(S_T)$ is the payoff function of the option.

The Case that $f(S_T) \equiv 1$

Suppose we have zero-coupon bond with maturity T and face value 1.

$$C = \mathbb{E}_{\mathbb{P}}[M_T e^{-rT}] = e^{-rT} \Rightarrow \mathbb{E}_{\mathbb{P}}[M_T] = 1.$$

Then M_T can serve as the Radon-Nikodym derivative that induces a new probability measure $\mathbb{Q}$ with respect to the real-world measure $\mathbb{P}$, i.e.,

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = M_T,$$

i.e., for any event A, $\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}[M_T 1_A]$.

Under this new probability measure $\mathbb{Q}$, the option pricing formula becomes:

$$C = \mathbb{E}_{\mathbb{P}}[M_T e^{-rT} f(S_T)] = \mathbb{E}_{\mathbb{Q}}[e^{-rT} f(S_T)].$$

$$C = \int M_T(y) \cdot e^{-rT} f(y) \cdot \mathbb{P}(S_T \in dy)$$

$$Q(S_T \in dy) = E_{\mathbb{P}}\left[M_T \cdot \mathbb{1}_{\{S_T \in dy\}}\right] = M_T(y) \cdot E_{\mathbb{P}}\left[\mathbb{1}_{\{S_T \in dy\}}\right]$$

$$= M_T(y) \cdot \mathbb{P}(S_T \in dy)$$

$$C = \int e^{-rT} f(y)\, Q(S_T \in dy)$$

$$= E_Q\left[e^{-rT} f(S_T)\right]$$

The Case that $f(S_T) = S_T - K$ (a Forward Contract)

This is the case of holding a stock.

The contract value is given by:

$$C = \mathbb{E}_{\mathbb{P}}[e^{-rT} M_T (S_T - K)] = 0.$$

Therefore, the fair forward price is given by:

$$F_0 = \mathbb{E}_{\mathbb{P}}[M_T S_T].$$

The Fair Forward Price

In the case of no dividend yield, by the no-arbitrage principle, the fair forward price is also given by:

$$F_0 = S_0 e^{rT},$$

by the no-arbitrage arguments as follows: $F_0 - S_0 e^{rT} > 0$

- If $F_0 > S_0 e^{rT}$, sell a forward contract and buy the underlying asset by borrowing S_0 at the risk-free rate.
- If $F_0 < S_0 e^{rT}$, sell the underlying asset and buy a forward contract, and invest S_0 at the risk-free rate. $S_0 e^{rT} - F_0 > 0$

Therefore, we have:

$$E_{\mathbb{Q}}[S_T] = \mathbb{E}_{\mathbb{P}}[M_T S_T] = S_0 e^{rT}. \quad E_{\mathbb{P}}[M_T] = 1$$

The Black-Scholes Model

The Black-Scholes model specifies the underlying asset price S_T as:

$$S_T = S_0 e^{(\mu - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z},$$

where Z is a standard normal random variable.

The pricing factor M_T is given by:

$$M_T = e^{-\frac{1}{2}\theta^2 T - \theta\sqrt{T}Z},$$

for some $\theta \in \mathbb{R}$.

Noting that $\mathbb{E}_{\mathbb{P}}[M_T e^{-rT} S_T] = S_0$, we have

$$\theta = (\mu - r)/\sigma$$

which is known as the market price of risk.

The Distribution of S_T under the Risk-Neutral Measure

We note that S_T is a function of Z, which is a standard normal random variable under the real-world measure $\mathbb{P}$.

To see its distribution under the risk-neutral measure $\mathbb{Q}$, we consider its moment generating function:

$$\mathbb{E}_{\mathbb{Q}}[e^{uZ}] = \mathbb{E}_{\mathbb{P}}[e^{uZ} M_T] = \mathbb{E}_{\mathbb{P}}[e^{-\frac{1}{2}\theta^2 T + (u-\theta\sqrt{T})Z}]$$
$$= e^{-\frac{1}{2}\theta^2 T + \frac{1}{2}(u-\theta\sqrt{T})^2} = e^{\frac{1}{2}u^2 - u\theta\sqrt{T}}.$$

The right-hand side is the moment generating function of a normal random variable with mean $-\theta\sqrt{T}$ and variance 1. Then,

$$\widetilde{Z} = Z + \theta\sqrt{T} \sim \mathcal{N}(0,1), \text{ under } \mathbb{Q}.$$

And,

$$S_T = S_0 e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\widetilde{Z}}.$$

The Black-Scholes Formula

The Black-Scholes formula for European call option price is given by:

$$C = \mathbb{E}_{\mathbb{Q}}[e^{-rT}(S_T - K)^+] = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2),$$

where Φ is the cumulative distribution function of a standard normal random variable, and

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

The Black-Scholes formula for European put option price is given by:

$$P = \mathbb{E}_{\mathbb{Q}}[e^{-rT}(K - S_T)^+] = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1).$$

$$C = E_Q\left[e^{-rT} \cdot \max\left(S_0 e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,\tilde{z}} - K,\ 0 \right) \right]$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \max\left(S_0 e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,z} - K,\ 0 \right) dz$$

$$z > z^* = \frac{\ln(K/S_0) - (r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$= \int_{z^*}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \cdot \left(S_0 e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,z} - K \right) dz$$

$$= \int_{z^*}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \cdot S_0 e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}z}\, dz - K\int_{z^*}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\, dz$$

The Case of Dividend Yield

For simplicity, we consider constant proportional dividend yield q.

The pricing factor M_T should satisfy that:

$$\mathbb{E}_{\mathbb{P}}[M_T e^{-rT} e^{qT} S_T] = S_0, M_T = e^{-\frac{1}{2}\theta^2 T - \theta\sqrt{T}Z},$$

$\theta = (\mu - (r-q))/\sigma.$

In this case,

$$S_T = S_0 e^{(r-q-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\widetilde{Z}}, \quad \mathbb{E}_{\mathbb{Q}}[e^{-rT} S_T] = S_0 e^{-qT}.$$

The Black-Scholes Model with Dividend Yield

The Black-Scholes formula for European call option price with dividend yield is given by:

$$C = \mathbb{E}_{\mathbb{Q}}[e^{-rT}(S_T - K)^+] = S_0 e^{-qT}\Phi(d_1) - Ke^{-rT}\Phi(d_2),$$

where $d_1 = \frac{\ln(S_0/K)+(r-q+\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, and $d_2 = d_1 - \sigma\sqrt{T}$.

The Black-Scholes formula for European put option price with dividend yield is given by:

$$P = \mathbb{E}_{\mathbb{Q}}[e^{-rT}(K - S_T)^+] = Ke^{-rT}\Phi(-d_2) - S_0 e^{-qT}\Phi(-d_1).$$

The General Dividend Yield Case

In practice, dividends are not distributed continuously, but rather at discrete times. Moreover, it is not guaranteed that the dividend yield is constant.

$F_0 = \mathbb{E}_{\mathbb{Q}}[S_T]$

In this case, we may recognize that the forward price process satisfies that

$$\mathbb{E}_{\mathbb{Q}}[F_T] = F_0, \qquad F_T = S_T.$$

Therefore, we may specify that

$$F_T = F_0 e^{-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}\widetilde{Z}}.$$

BS model for the forward price

The Black-Scholes Formula by the Forward Price

And then the Black-Scholes formula for European call option price with dividend yield is given by:

$$C = \mathbb{E}_{\mathbb{Q}}[e^{-rT}(F_T - K)^+] = e^{-rT}(F_0\Phi(d_1) - K\Phi(d_2)),$$

where $d_1 = \frac{\ln(F_0/K)+\frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$, and $d_2 = d_1 - \sigma\sqrt{T}$.

The same argument applies to put options with formula:

$$P = \mathbb{E}_{\mathbb{Q}}[e^{-rT}(K - F_T)^+] = e^{-rT}(K\Phi(-d_2) - F_0\Phi(-d_1)).$$

In the continuous dividend yield case, the forward price is given by:

$$F_0 = S_0 e^{(r-q)T}.$$

The Put-Call Parity

The put-call parity is given by: $F_0 = e^{rT}(C-P) + K$

$$C - P = e^{-rT}(F_0 - K).$$

If it is violated, there exists an arbitrage opportunity as follows:

- If $C - P > e^{-rT}(F_0 - K)$, sell a call option and buy a put option, and enter a forward contract to buy the underlying asset at the forward price F_0. $[C-P]\cdot e^{rT} - \max(S_T - K, 0) + \max(K - S_T, 0) + S_T - F_0$
- If $C - P < e^{-rT}(F_0 - K)$, buy a call option and sell a put option, and enter a forward contract to sell the underlying asset at the forward price F_0.

$S_T > K: [C-P]e^{rT} - (F_0 - K) \geq 0$

$S_T < K: [C-P]e^{rT} - (F_0 - K) \geq 0$

Calibrating the Black-Scholes Model to Option Data: Implied Volatility

There is only one free parameter in the Black-Scholes model, which is the volatility σ.

The implied volatility is the volatility that makes the Black-Scholes formula fit the option price, i.e.,

$$C_{\mathrm{BS}}(K, T; \sigma) = C_{\mathrm{MKT}}(K, T),$$

where C_{BS} is the Black-Scholes formula, and C_{MKT} is the market price of the option.

This yields a nonlinear equation for σ, which can be solved by numerical methods:

$$\sigma = \sigma_{\mathrm{IV}}(K, T).$$

The Implied Volatility Surface

It is well-known that the implied volatility surface is not flat, but rather exhibits a smile or skew pattern.

- Implied volatility curve: the dependence of implied volatility on the strike price at a fixed maturity.
- Term structure of implied volatility: the dependence of implied volatility on the maturity date.

These dependence violates the Black-Scholes model, which assumes that the implied volatility is constant over time and strike prices.

However, implied volatility calculated by the Black-Scholes formula serves as a convenient measure of the highness of the option price.

Existence of Implied Volatility

By the no-arbitrage principle, the market price of a call option must belong to a range:

$$(S_0 - Ke^{-rT})^+ \leq C_{\text{MKT}} \leq S_0.$$

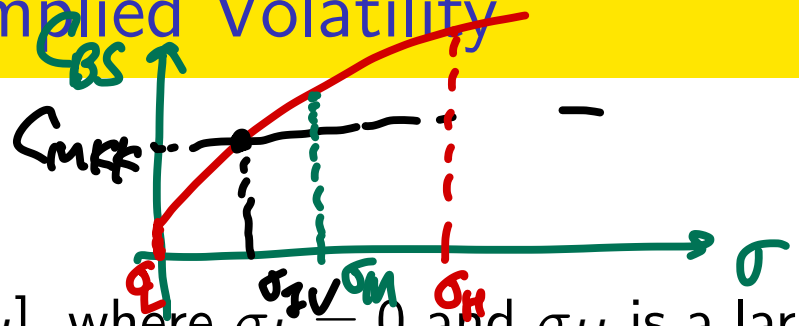
Moreover, we note that

$$\lim_{\sigma \to 0} C_{\text{BS}}(K, T; \sigma) = (S_0 - Ke^{-rT})^+,$$
$$\lim_{\sigma \to \infty} C_{\text{BS}}(K, T; \sigma) = S_0.$$

Therefore, the implied volatility must exist if the market price of the option belongs to the no-arbitrage range. The same argument applies to put options.

Numerical Methods for Implied Volatility

The bisection search method.



1. Set the initial range $[\sigma_L, \sigma_H]$, where $\sigma_L = 0$ and σ_H is a large number.
2. Calculate $\sigma_M = \frac{\sigma_L + \sigma_H}{2}$ and evaluate $C_{\text{BS}}(K, T; \sigma_M)$.
3. If $|C_{\text{BS}}(K, T; \sigma_M) - C_{\text{MKT}}| < \epsilon$, stop.
4. Otherwise:
   - If $C_{\text{BS}}(K, T; \sigma_M) > C_{\text{MKT}}$, set $\sigma_H = \sigma_M$
   - If $C_{\text{BS}}(K, T; \sigma_M) < C_{\text{MKT}}$, set $\sigma_L = \sigma_M$
5. Return to step 2.

$$(\sigma_H - \sigma_L) \cdot 2^{-n} = \epsilon$$

$$n = \log_2 \frac{\sigma_H - \sigma_L}{\epsilon}$$

The number of iterations to achieve the desired accuracy ϵ is given by:

$$\log_2 \left(\frac{\sigma_H - \sigma_L}{\epsilon} \right).$$

The Newton-Raphson Method

The Newton-Raphson method is a more efficient method for finding the implied volatility.

The iteration of the Newton-Raphson method is given by:

$$\sigma_{n+1} = \sigma_n - \frac{C_{\text{BS}}(K, T; \sigma_n) - C_{\text{MKT}}}{\partial_\sigma C_{\text{BS}}(K, T; \sigma_n)}.$$

Vega

The Newton-Raphson method converges faster than the bisection search method.

SPX Options

The standard options written on the S&P 500 index are European style. The liquidity of SPX options is very high, and the data is publicly available, providing a good testbed for developing and testing option pricing models.

Apart from option pricing and trading, options embeds a lot of forward looking information about the market.

We are concerned about the following issues:

- How to estimate the implied volatility of SPX options?
- How to extract volatility, skewness and kurtosis from the option prices?
- How to construct a distribution model for S_T under the risk-neutral measure to explain the implied volatility curve?

The SPX Options Data

The December 13, 2024 trading data of SPX options expiring on January 17, 2025.

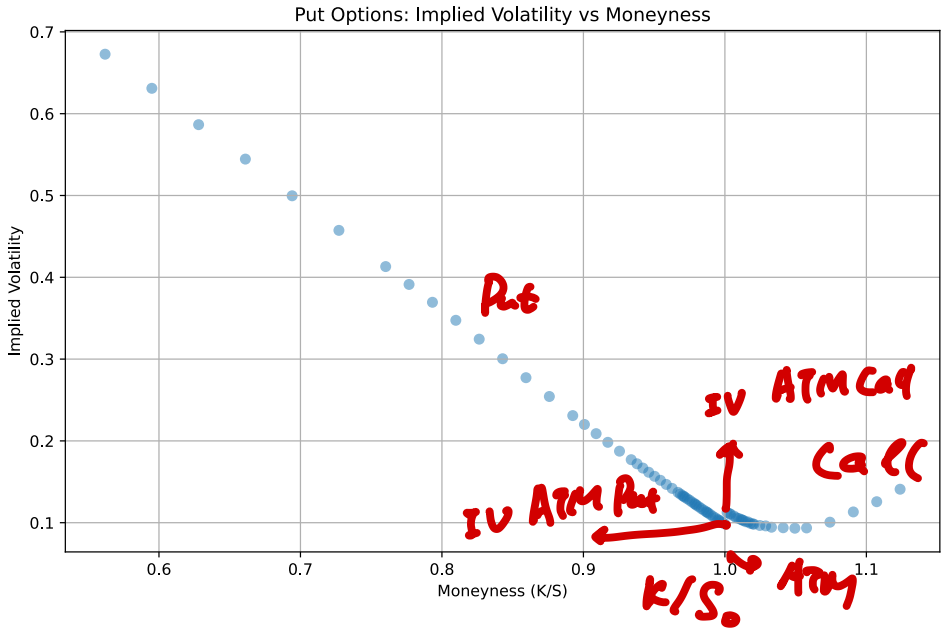
The current S&P 500 index level is 6051, the 3-month risk-free rate is 4.31% per annum, and the dividend yield is 1.213% per annum.

Strike	Moneyness	Bid	Mid	Ask	Volume	Open Int	Delta	IV	Type
2600	-0.5703	0.00	0.13	0.25	0	0	-0.00030	0.8661	Put
2800	-0.5373	0.00	0.15	0.30	0	0	-0.00038	0.8045	Put
3000	-0.5042	0.05	0.20	0.35	0	0	-0.00053	0.7527	Put
3200	-0.4712	0.05	0.23	0.40	0	0	-0.00064	0.6939	Put
3400	-0.4381	0.15	0.33	0.50	0	94	-0.00096	0.6515	Put
7000	0.1568	907.6	917.55	927.5	2	0	0.0	0.0	Call
7200	0.1898	1106.6	1116.70	1126.8	0	0	0.0	0.0	Call
7400	0.2229	1305.6	1315.80	1326.0	0	0	0.0	0.0	Call
7600	0.2559	1504.6	1515.45	1526.3	0	0	0.0	0.0	Call
7800	0.2890	1703.6	1714.55	1725.5	0	0	0.0	0.0	Call

Table: First and Last Few Rows of SPX Options Data.

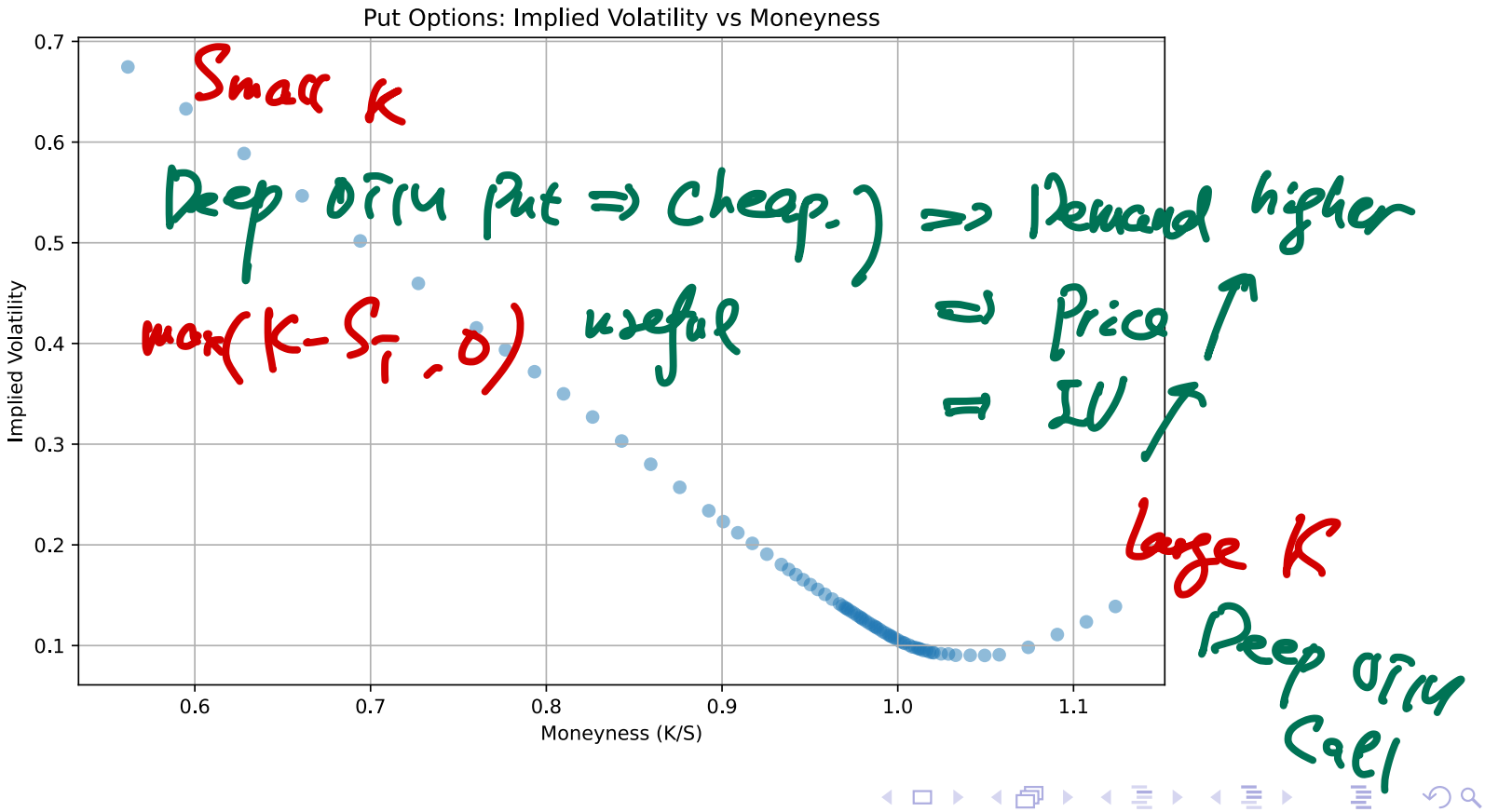
The Implied Volatility Curve

We calculate the implied volatility by OTM SPX call (converted to put options by the put-call parity) and put options (filtering out options with zero volume and open interest) by inverting the Black-Scholes formula assuming the dividend yield is 1.213% per annum. We see inconsistency of the implied volatility implied from OTM put options and OTM call options .



Correction by the Implied Forward Price

We first calculate the implied forward price by the put-call parity using ATM call and put options, and then use the implied forward price to calculate the implied volatility.



The Volatility Index (VIX)

$C(K) = E_Q[e^{-rT}\max(S_T - K, 0)]$

$P(K) \Rightarrow E_Q[e^{-rT}\max(K - S_T, 0)]$

Dist of S_T under Q $\Leftrightarrow$ $C(K)$, $P(K)$ for all K

VIX is a volatility index that is widely used to measure the market's expectation of volatility over the next 30 days. It is releaseed by the CBOE (Chicago Board Options Exchange) every day at 4pm EST from 2004.

The CBOE VIX index can be derived from the prices of SPX options using the following formula:

Put option

Call option

$$\text{VIX} = 100 \times \sqrt{\frac{2e^{rT}}{T}\left(\int_0^{F_0} \frac{1}{K^2}P(K)dK + \int_{F_0}^{\infty} \frac{1}{K^2}C(K)dK\right)},$$

where F_0 is the forward price, $P(K)$ and $C(K)$ are put and call option prices, and r is the risk-free rate.

$\int_0^\infty \frac{1}{K^2} Q(K)\, dK$

$$C(K) = E_Q\left[e^{-rT}\max(S_T - K, 0)\right]$$ → $f(x)$ Density of S_T under Q

$$= e^{-rT}\int_{-\infty}^{\infty} \max(x-K, 0)\cdot f(x)\,dx$$

$$= e^{-rT}\int_{K}^{\infty} (x-K)\, f(x)\,dx$$

$$= e^{-rT}\int_{K}^{\infty} x\, f(x)\,dx - e^{-rT}K\int_{K}^{\infty} f(x)\,dx$$

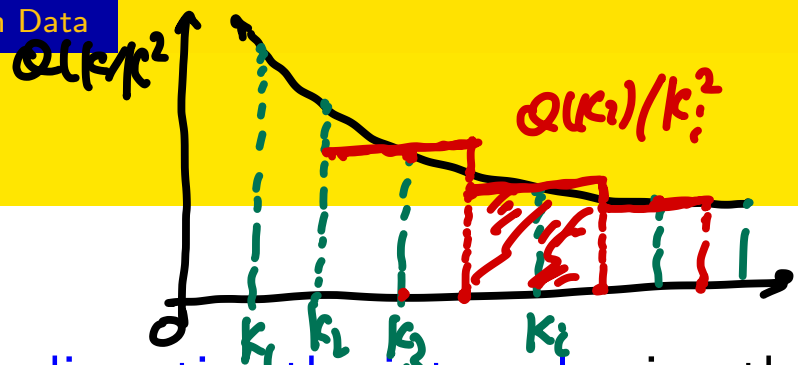
$$C'(K) = \cancel{e^{-rT}(-Kf(K))} - e^{-rT}\cdot\int_{K}^{\infty} f(x)\,dx$$

$$\cancel{- e^{-rT}K\cdot(-f(K))}$$

$$C''(K) = -e^{-rT} \cdot (-f(K)) = e^{-rT} f(K)$$

$$f(K) = e^{rT} C''(K)$$

Calculating the VIX



To calculate the VIX, we need to discretize the integral using the mid-point rule, which leads to the following formula:

$$\text{VIX} = 100 \times \sqrt{\frac{2e^{rT}}{T}\sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T}\left(\frac{F_0}{K_0} - 1\right)^2},$$

where $Q(K_i)$ is the midpoint of the bid-ask spread for each option with strike K_i, $\Delta K_i = (K_{i+1} - K_{i-1})/2$, and K_0 is the first strike below the forward price.

Using the SPX options data, we calculate the VIX to be 13.78 while the reported VIX is 13.81 (close value).

Implied Skewness and Kurtosis

Bakshi (2003)[9] propose methods to extract implied skewness and kurtosis of $\ln(S_T/S_0)$ from the option prices.

$$\text{SKEW}(T) = \frac{e^{rT}W(T) - 3\mu(T)e^{rT}V(T) + 2\mu(T)^3}{(e^{rT}V(T) - \mu(T)^2)^{3/2}},$$

$$\text{KURT}(T) = \frac{e^{rT}X(T) - 4\mu(T)e^{rT}W(T) + 6e^{rT}\mu(T)^2V(T) - 3\mu(T)^4}{(e^{rT}V(T) - \mu(T)^2)^2},$$

where

$$V(T) = \int_0^\infty \frac{2(1 - \ln(K/S_0))}{K^2} Q(K)dK,$$

$$X(T) = \int_0^\infty \frac{12(\ln(K/S_0))^2 - 4(\ln(K/S_0))^3}{K^2} Q(K)dK,$$

$$W(T) = \int_0^\infty \frac{6\ln(K/S_0) - 3(\ln(K/S_0))^2}{K^2} Q(K)dK.$$

[9] Bakshi, Gurdip, Nikunj Kapadia, and Dilip Madan. "Stock return characteristics, skew laws, and the differential pricing of individual equity options." The Review of Financial Studies 16, no. 1 (2003): 101-143.

Implied Mean and Volatility

Also we have that the implied mean and standard deviation of the log return is given by

$$\mu(T) = e^{rT} - 1 - \frac{e^{rT}V}{2} - \frac{e^{rT}W}{6} - \frac{e^{rT}X}{24},$$
$$\sigma(T) = \sqrt{Ve^{rT} - \mu(T)^2}.$$

The results are as follows:

BKM Risk-Neutral Moments (30 days).	
Mean	0.0028
Volatility	0.0411
Skewness	−5.7389
Kurtosis	72.0314

Risk-neutral dist is much heavier than P- dist.

The Implied NIG Model

Suppose $\ln(S_T/S_0)$ follows a normal inverse Gaussian distribution, i.e.,

$$\ln(S_T/S_0) \sim NIG(\alpha, \beta, \delta, \mu).$$

The parameters can be estimated by matching the mean, variance, skewness and kurtosis with those implied by BKM (2003).

$$\mu(T) = \mu + \delta\beta/\gamma,$$
$$\sigma^2(T) = \delta\alpha^2/\gamma^3,$$
$$\text{SKEW}(T) = 3\beta/(\alpha\sqrt{\delta\gamma}),$$
$$\text{KURT}(T) = 3(1 + 4\beta^2/\alpha^2)/(\delta\gamma) + 3.$$

The Estimated NIG Parameters

The estimated NIG parameters are as follows:

Fitted NIG Parameters	
α	15.0643
β	-9.9777
δ	0.0106
μ	0.0121

$|\beta| < |\alpha|$

We check the moments of the NIG distribution and compare them with those implied by BKM (2003).

Moments Comparison		
	BKM	NIG
Mean	0.0028	0.0028
Volatility	0.0411	0.0409
Skewness	-5.7389	-5.7389
Kurtosis	72.0314	71.9375

The Performance of the Implied NIG Model

We examine the performance of the implied NIG model by comparing the implied option prices and implied volatility with the observed.

The option prices implied by the NIG model are estimated by Monte Carlo simulation:

$$P_{\text{NIG}}(K,T) = \frac{1}{N}\sum_{i=1}^{N} e^{-rT}\max(0, S_0 e^{X_i} - K),$$

where X_i is simulated from the fitted NIG distribution and $N = 100000$ is the number of simulations.

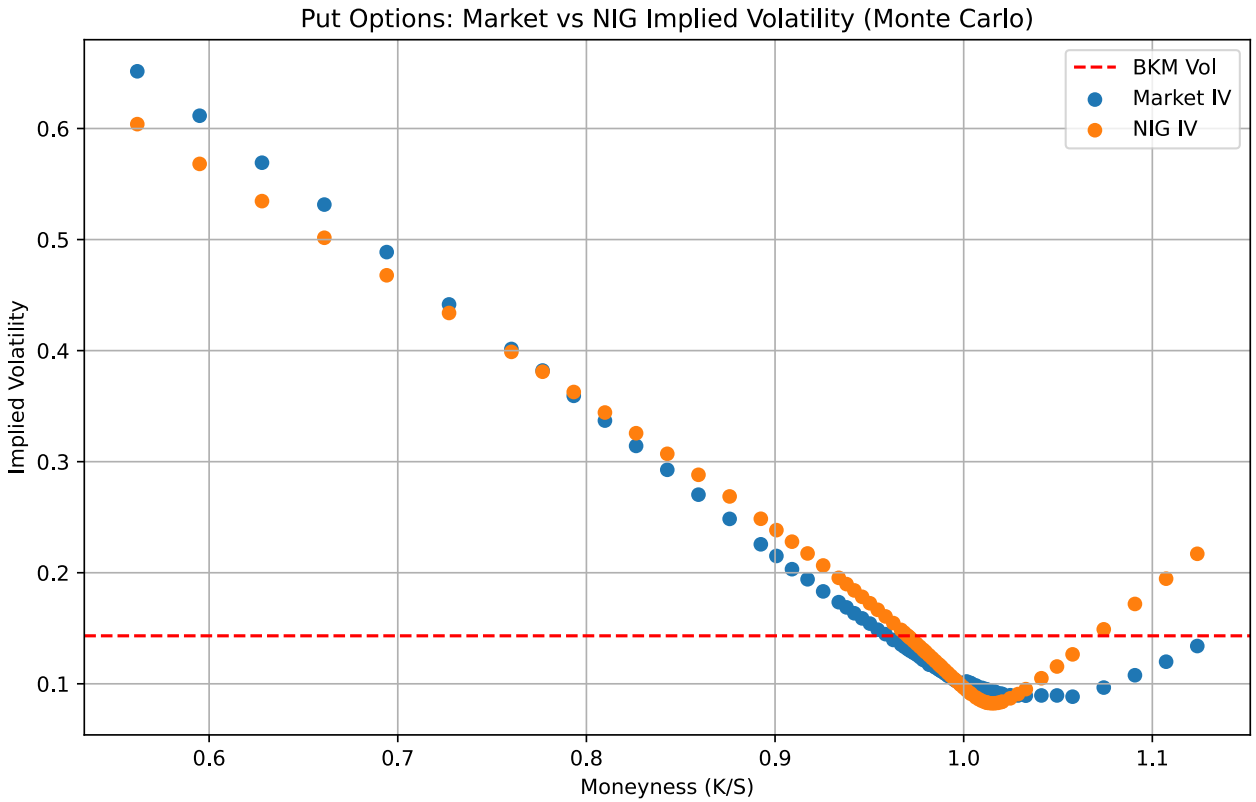
Errors for Different Moneyness Groups

Deep OTM ($K/S < 0.94$), OTM ($0.94 \leq K/S < 0.97$), ATM ($0.97 \leq K/S < 1.03$), ITM ($1.03 \leq K/S < 1.06$), Deep ITM ($K/S \geq 1.06$).

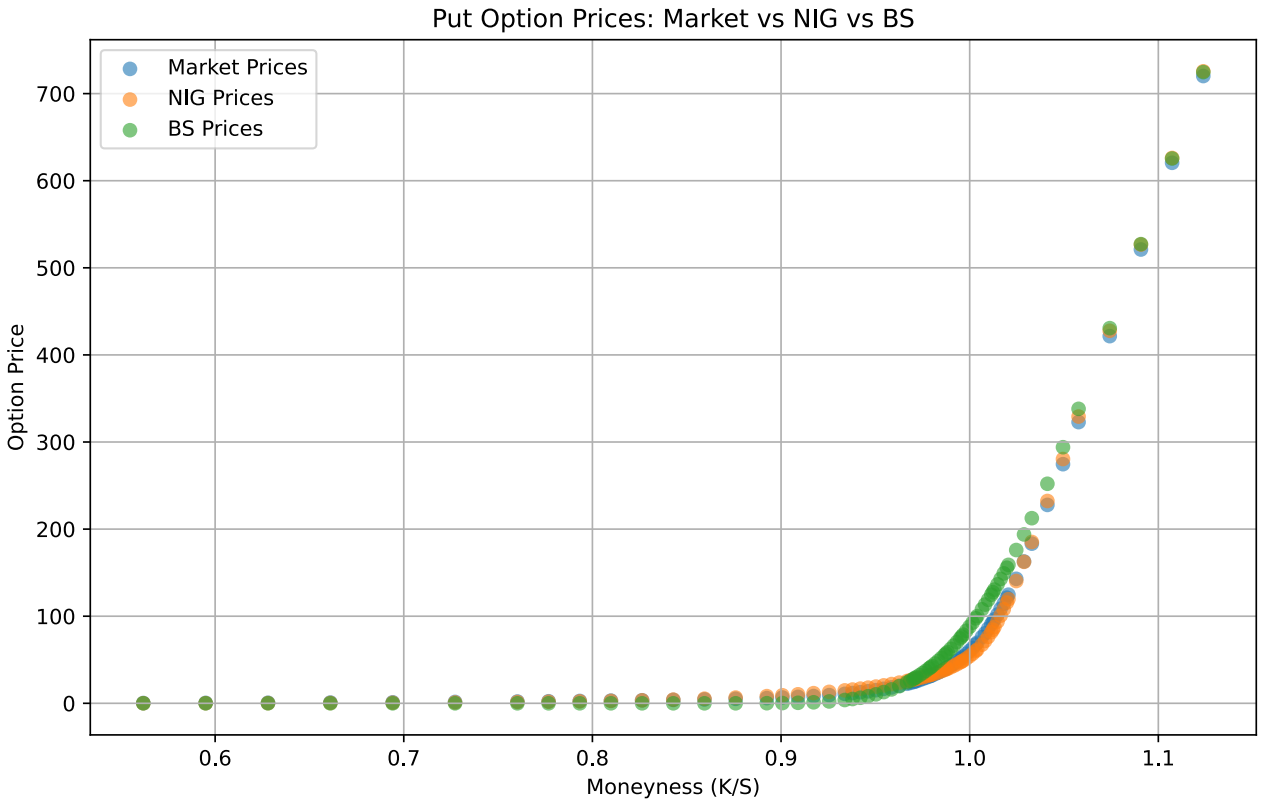
ITM just converted from OTM call by put call parity

	NIG Model		BS Model	
K/S	Rel. Error	Abs. Error	Rel. Error	Abs. Error
< 0.94	32.76%	1.46	93.82%	3.79
$[0.94, 0.97)$	23.15%	3.86	22.45%	3.45
$[0.97, 1.03)$	6.84%	4.32	35.16%	22.16
$[1.03, 1.06)$	1.66%	4.36	9.56%	21.95
≥ 1.06	0.99%	5.38	1.23%	6.41
Overall	15.14%	3.54	46.90%	14.27

NIG Implied Volatility Curve



Market vs NIG and BS Prices



To Refine the NIG Model

We can refine the parameters of the NIG model by minizing the following objective function:

$$\min_{\alpha,\beta,\delta,\mu} \frac{1}{n}\sum_{i=1}^{n}\left(P_{\text{NIG}}(K_i, T) - P_{\text{MKT}}(K_i, T)\right)^2,$$

where $P_{\text{NIG}}(K_i, T)$ is the price implied by the NIG model and $P_{\text{MKT}}(K_i, T)$ is the market price of the option with strike K_i and maturity T.

Here the $P_{\text{NIG}}(K_i, T)$ is calculated by Monte Carlo simulation.

Additional penalty terms can be added to the objective function to ensure the validity of the NIG parameters ($\delta > 0$ and $|\alpha| > |\beta|$) and the no-arbitrage bounds of the option prices:

$$e^{-rT}(K - F_0)^+ \leq P_{\text{NIG}}(K, T) \leq e^{-rT}K.$$

Simulating NIG Random Variables

To avoid repeated sampling for different parameters, we sample the NIG distribution as follows:

$$X = \mu + \delta(bx + \sqrt{x}W),$$

where $b = \beta\delta$, W is a standard normal random variable and x is sampled from the inverse Gaussian distribution as follows:

$$y = \mu_{ig} + \frac{1}{2}\mu_{ig}^2 Z^2 - \frac{1}{2}\mu_{ig}\sqrt{4\mu_{ig}Z^2 + \mu_{ig}^2 Z^4},$$

$$x = \begin{cases} y & \text{if } U \leq \frac{\mu_{ig}}{\mu_{ig}+y}, \\ \frac{\mu_{ig}^2}{y} & \text{otherwise.} \end{cases}$$

where $a = \alpha\delta$, $\mu_{ig} = 1/\sqrt{a^2 - b^2}$, U is sampled from the uniform distribution on $[0, 1]$ and Z is sampled from the standard normal distribution.

The Estimated NIG Parameters

The parameters:

Parameter	Value
α	20.4877
β	-9.8971
δ	0.0188
μ	0.0142

The statistics of the fitted NIG distribution:

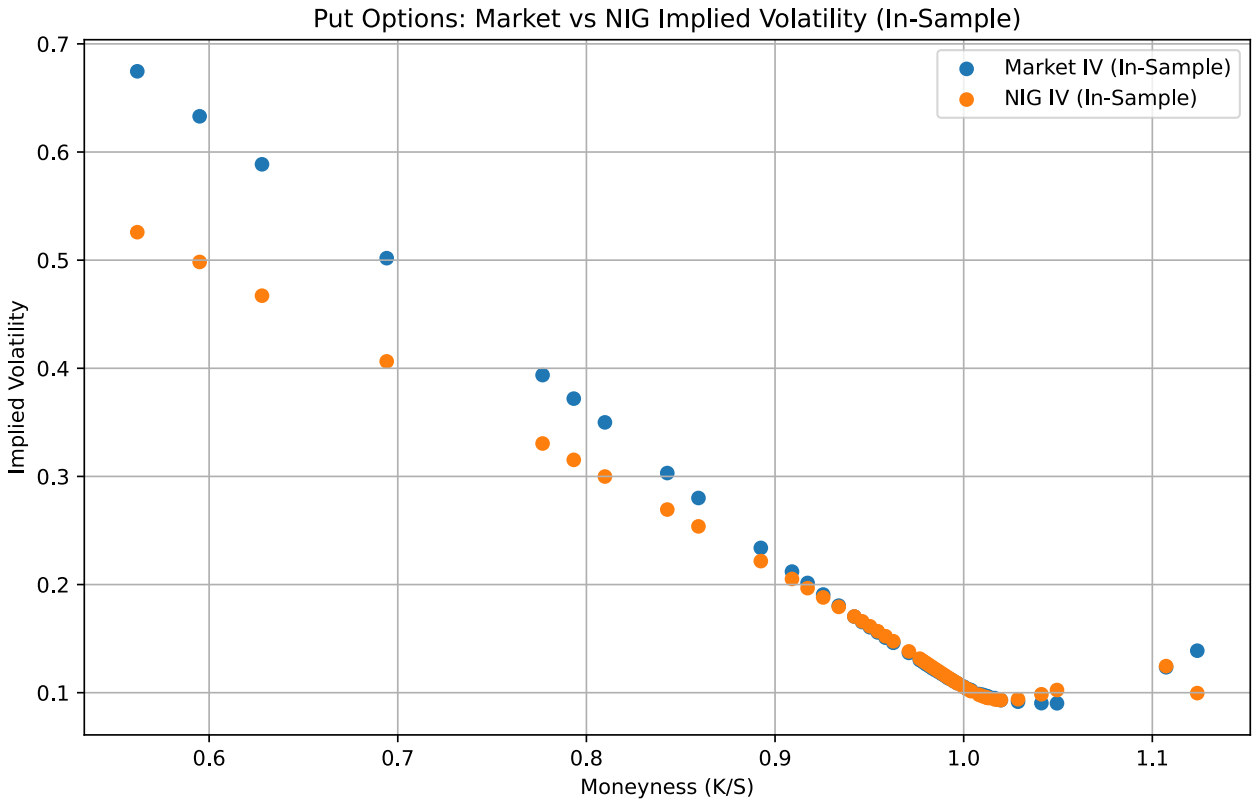
Moment	Value
Mean	0.0038
Volatility	0.0369
Skewness	-2.4972
Kurtosis	20.2219

In-Sample and Out-of-Sample Performance

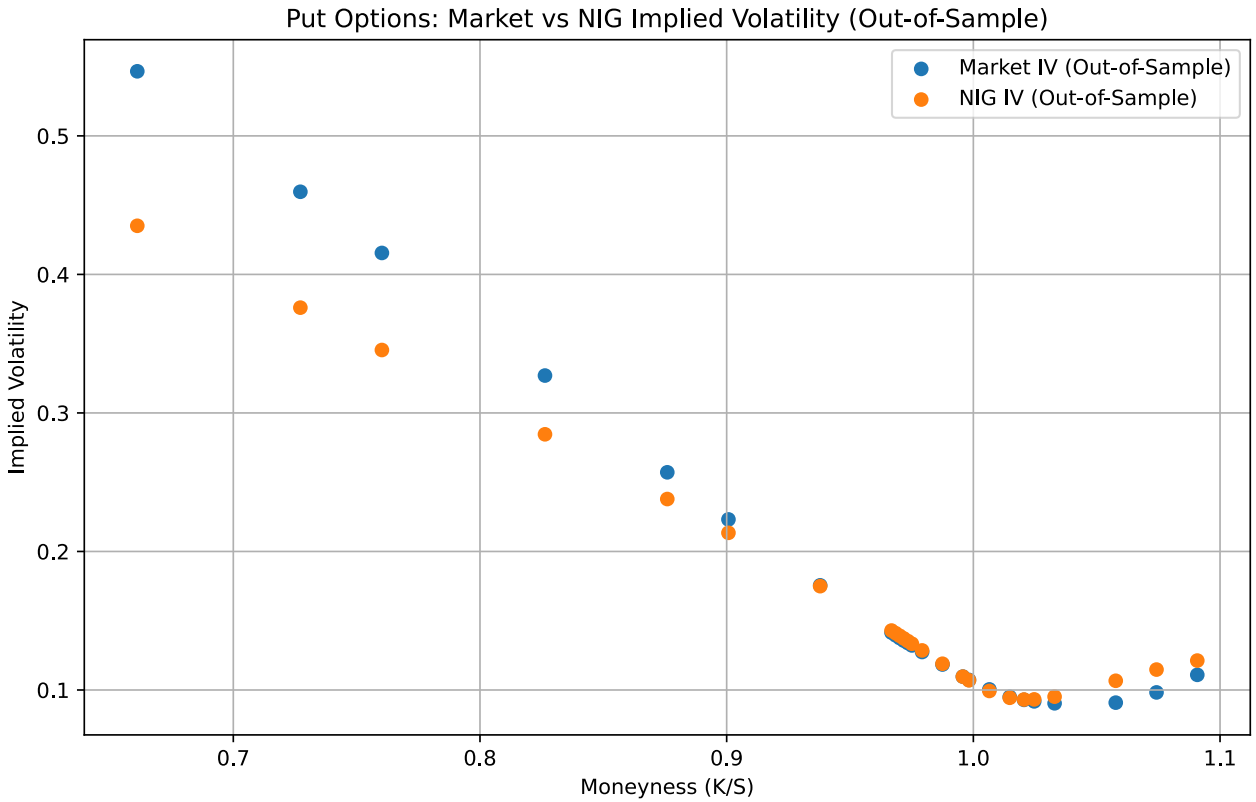
We randomly split the data into in-sample (approximately 70%) and out-of-sample (approximately 30%). The parameters are estimated using the in-sample data.

	In-Sample		Out-of-Sample	
K/S	Rel. Error	Abs. Error	Rel. Error	Abs. Error
< 0.94	54.59%	1.23	55.02%	1.37
$[0.94, 0.97)$	1.93%	0.33	2.91%	0.67
$[0.97, 1.03)$	0.96%	0.50	1.33%	0.50
$[1.03, 1.06)$	0.97%	2.43	0.86%	2.03
≥ 1.06	0.01%	0.10	0.16%	0.70
Overall	15.76%	0.74	16.36%	0.90

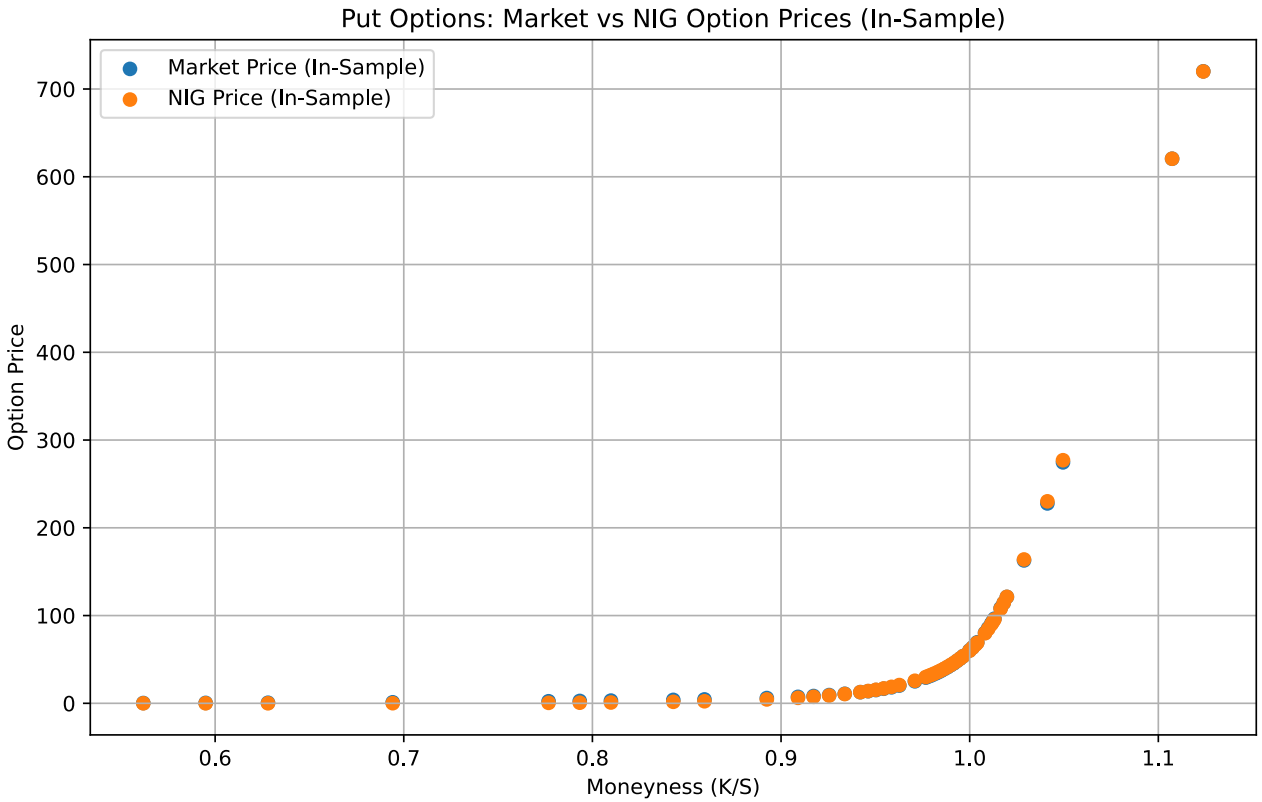
The NIG Implied Volatility Curve: In-Sample



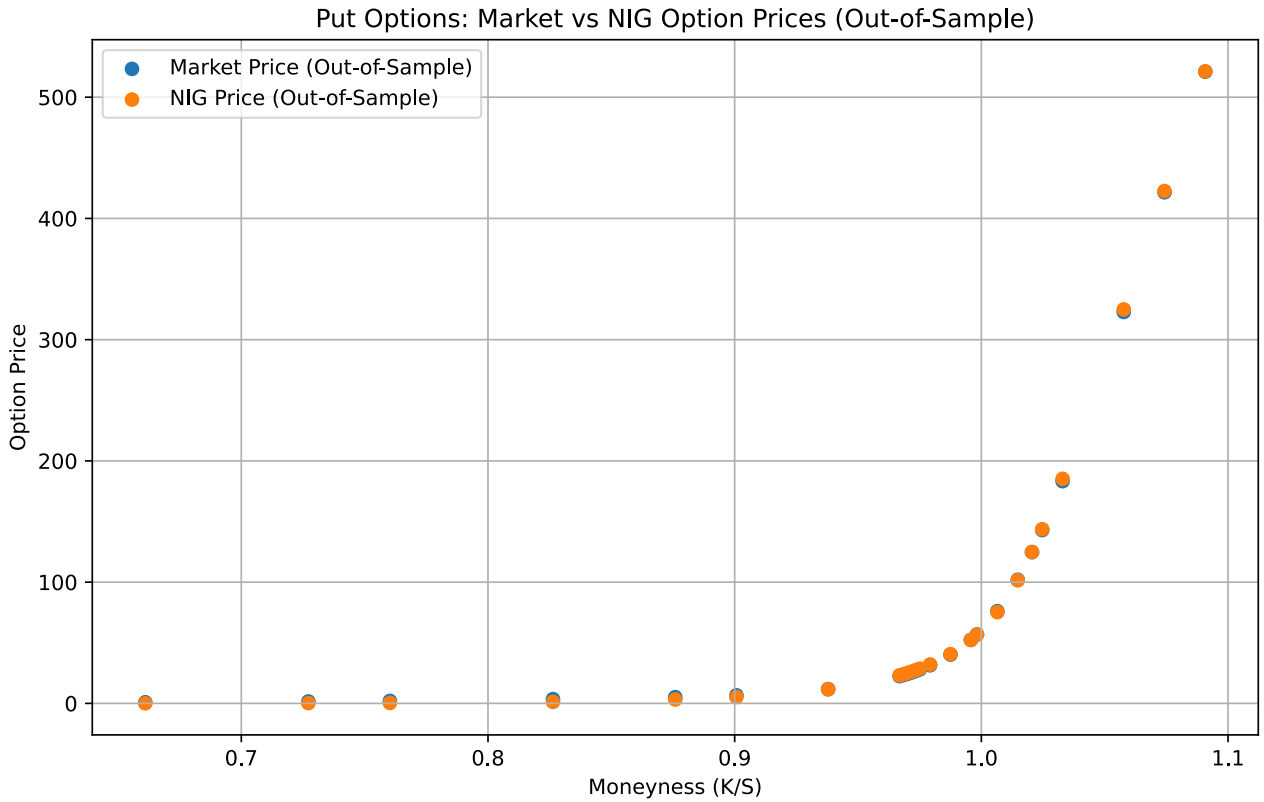
The NIG Implied Volatility Curve: Out-of-Sample



The NIG Price vs Market Price: In-Sample



The NIG Price vs Market Price: Out-of-Sample



The DEJD Model

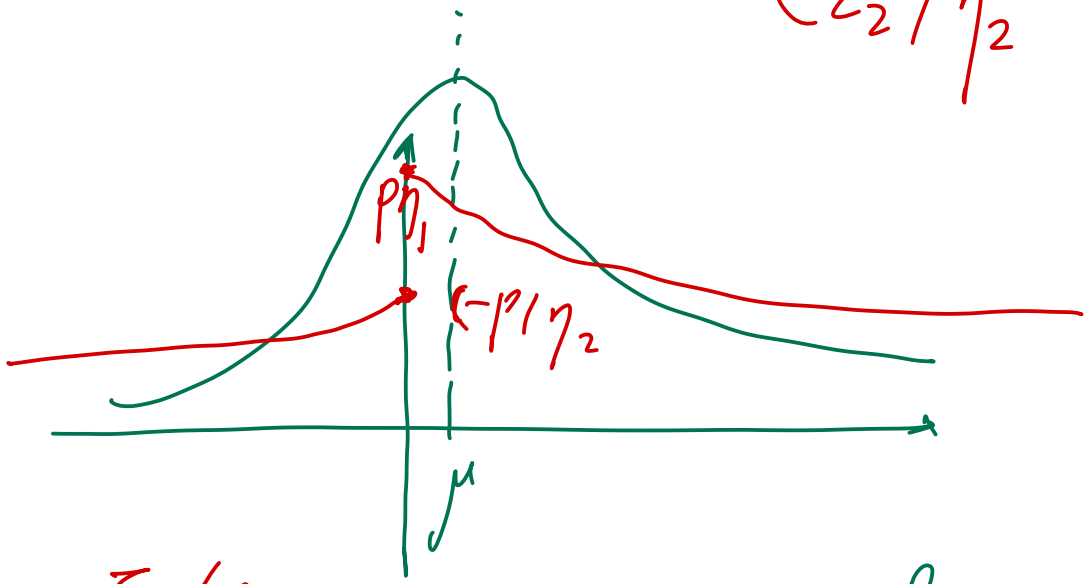
Here we consider a simple, intuitive and flexible model for the distribution of the log return, the DEJD (double exponential jump diffusion) model:

$$\ln(S_T/S_0) = \mu + \sigma\tilde{Z} + \begin{cases} \mathcal{E}_1/\eta_1 & \text{w.p. } p\lambda, \\ -\mathcal{E}_2/\eta_2 & \text{w.p. } (1-p)\lambda, \end{cases}$$

where $\tilde{Z}$ is a standard normal random variable, $\mathcal{E}_1$ and $\mathcal{E}_2$ are two independent exponential random variables with mean 1.

- λ is the probability of a jump.
- p is the probability of a positive jump given a jump occurs.
- $1/\eta_1$ and $1/\eta_2$ are the mean of the positive and negative jumps, respectively.

$$\ln\left(\frac{S_T}{S_0}\right) = \mu + \sigma\tilde{Z} + \begin{cases} \xi_1/\eta_1 & \text{w.p. } \lambda p \\ \xi_2/\eta_2 & \text{w.p } \lambda(1-p) \end{cases}$$



$$J = \begin{cases} \xi_1/\eta_1 & \text{w.p. } p \\ -\xi_2/\eta_2 & \text{w.p } 1-p \end{cases}$$

$$f(z) = \begin{cases} p\eta_1 e^{-\eta_1 z} & z \geq 0 \\ (1-p)\eta_2 e^{\eta_2 z} & z < 0 \end{cases}$$

The Estimated DEJD Parameters

Parameter	Value
μ	0.0048
σ	0.0251
λ	0.6195
p	0.9742
η_1	239.6988
η_2	3.5663

Frequent but small positive jumps

$1/\eta_1$

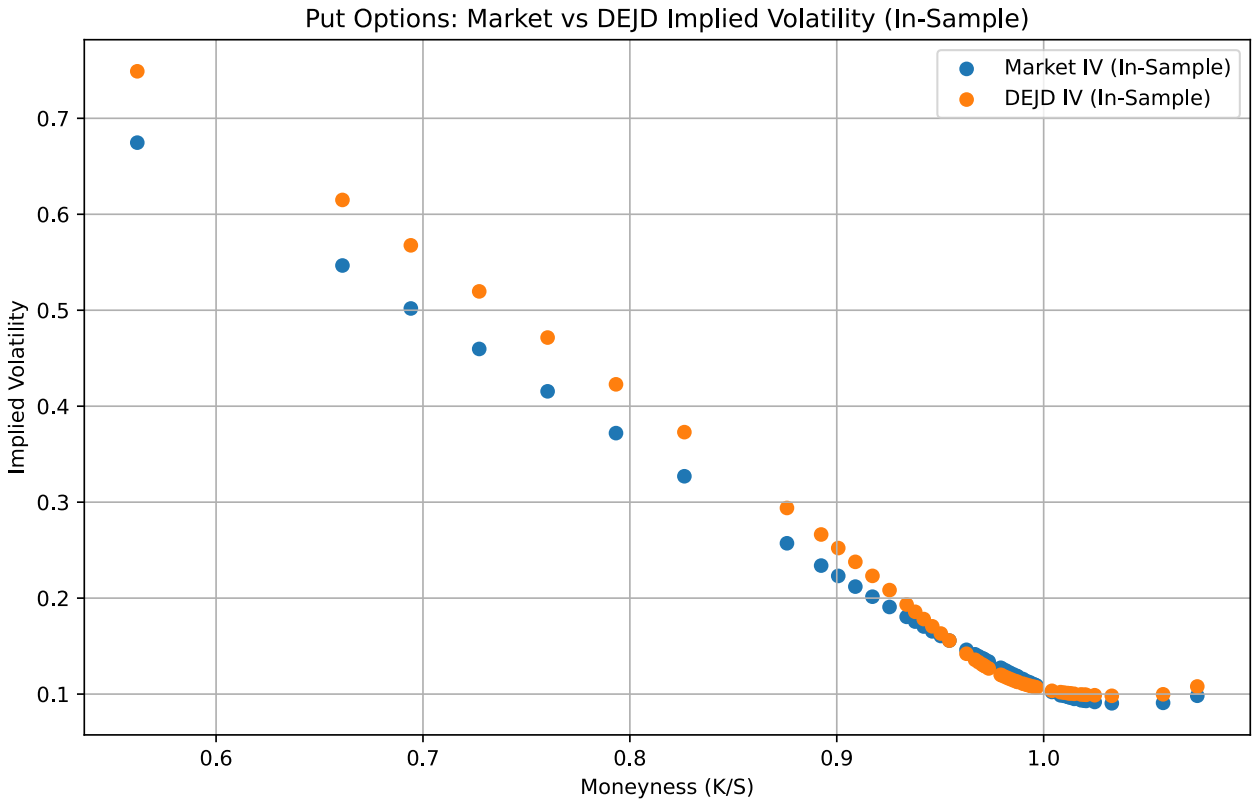
$1/\eta_2$

Rare but large negative jumps

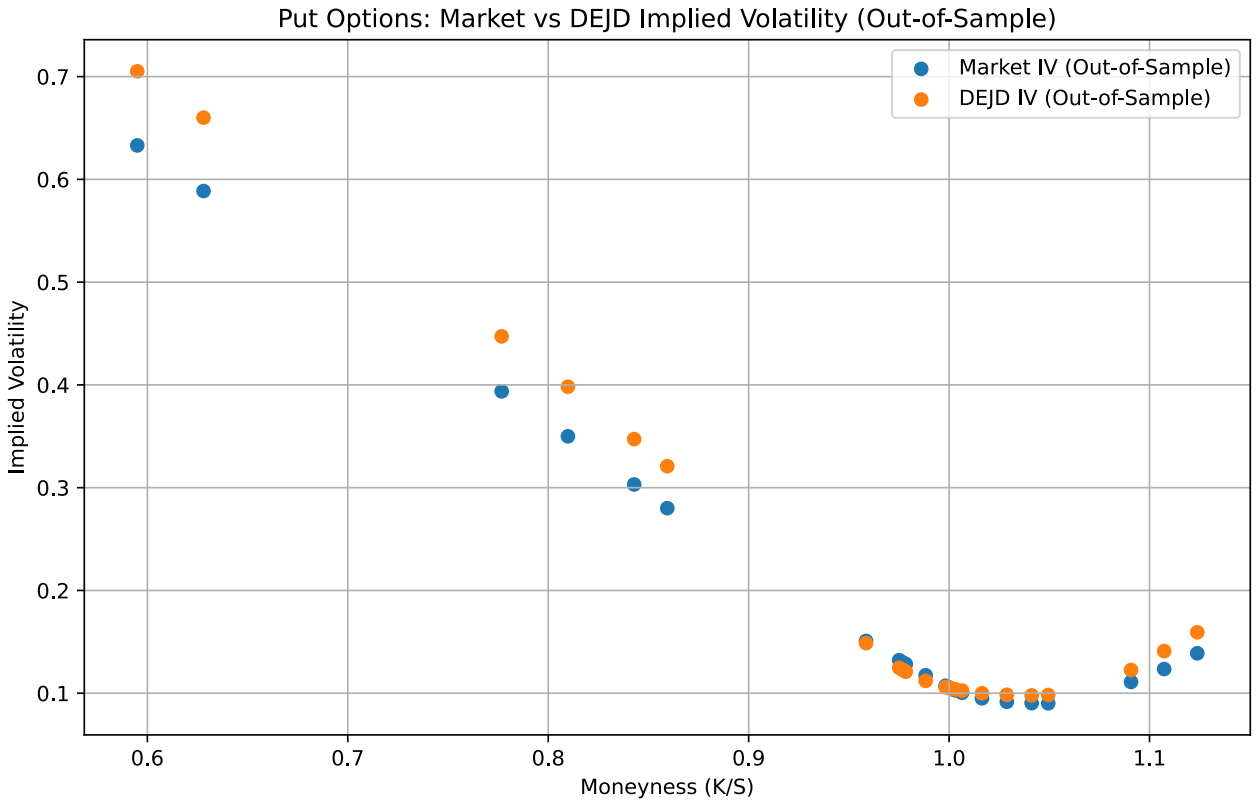
In-Sample and Out-of-Sample Performance

	In-Sample		Out-of-Sample	
K/S	Rel. Error	Abs. Error	Rel. Error	Abs. Error
< 0.94	109.18%	3.62	149.83%	3.24
$[0.94, 0.97)$	9.43%	1.65	4.39%	0.80
$[0.97, 1.03)$	6.43%	2.96	5.17%	2.24
$[1.03, 1.06)$	1.04%	2.13	0.76%	1.86
≥ 1.06	0.12%	0.52	0.07%	0.45
Overall	35.59%	2.90	41.83%	2.17

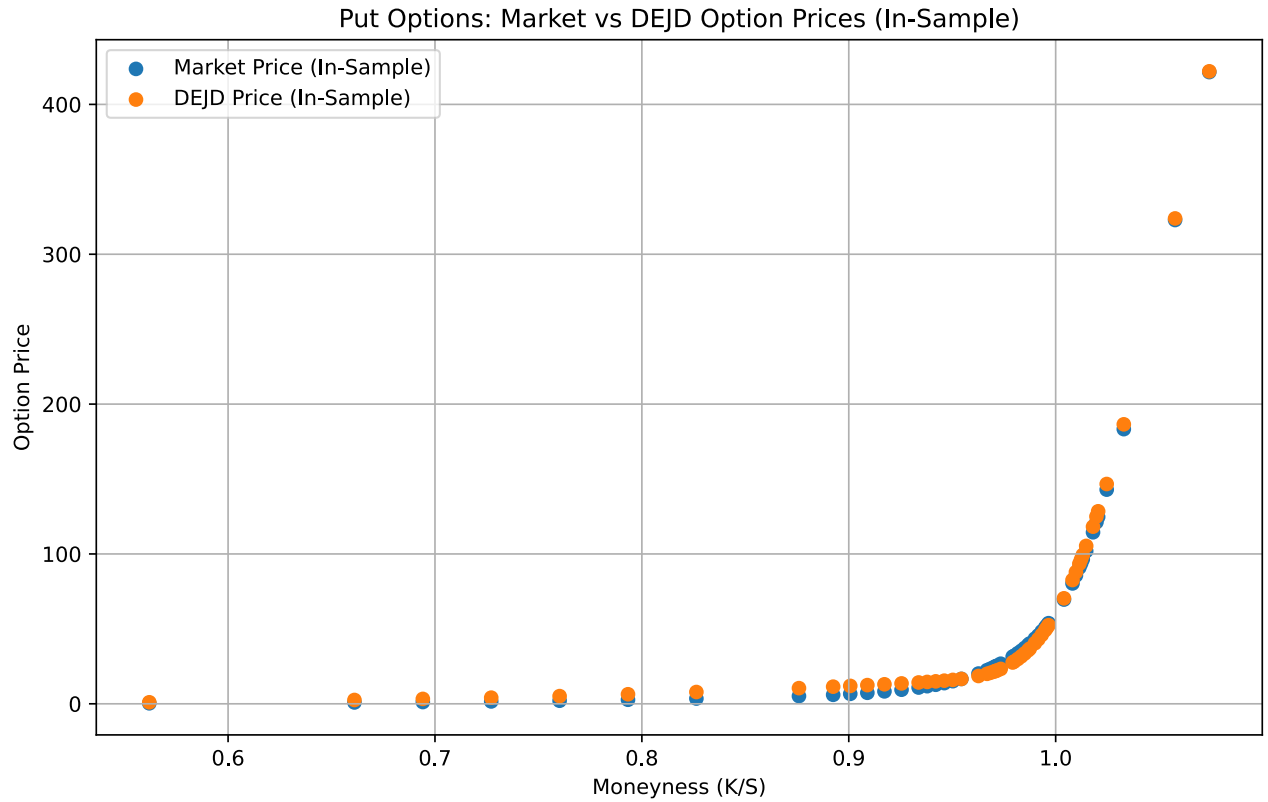
The DEJD Implied Volatility Curve: In-Sample



The DEJD Implied Volatility Curve: Out-of-Sample



The DEJD Price vs Market Price: In-Sample



The DEJD Price vs Market Price: Out-of-Sample

